

SHORT TERM INTERNSHIP PROJECT REPORT
ON
Heritage treasures: An in-depth analysis UNESCO

World heritage sites in tableau

TO
S.V University, TIRUPATI



BACHELOR OF COMPUTER APPLICATIONS
SUBMITTED BY

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TIRUPATI - 517501

2025 - 2026

Certificate

This is certifying that the project report **“Semester internship project report on Heritage treasures: An in-depth analysis**

UNESCO World heritage sites in tableau” are bona fide work done and Submitted by **“THEEPUNOORU MITHUN**

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”, In Partial Full Filament of the requirements for the award of the Degree of Computer Applications by the affiliated to SRI VENKATESWARA UNIVERSITY, TIRUPATI, during the Year 2025 – 2026

PROJECT GUIDE

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ACKNOWLEDGEMENT

We would like to express our gratitude to all those who have been instrumental in the presentation of this project.

We are very much grateful to **Mr. P. RAVI VENKATESH, B.E.**, Correspondent of **GAYATRI DEGREE COLLEGE**, for his great support.

We are very much grateful of **Mrs. P. LAKSHMI, M.A., Director** of **GAYATRI DEGREE COLLEGE**, for her great support.

We are thankful to **Dr. P. VENUGOPAL RAO, M.Sc, M.Phil, PhD, MBA, PRINCIPAL** of **GAYATRI DEGREE COLLEGE**, for giving her continuous support to our internship.

We are happy to express our gratitude to **Mr. P. MADHU SUDHAN, M.Sc. (Computer Science). Lecturer in Computers, GAYATRI DEGREE COLLEGE**, for his constant motivation and valuable support for completing the internship.

It gives us pleasure to acknowledge our indebtedness to **All Teaching Faculties** and we are very much thankful to **our parents** for their substantial moral support and encouragement in our studies.

Place: TIRUPATI

Date:

DECLARATION

We hereby declare that the Project Work Entitled **“Semester internship project report on Heritage treasures: An in-depth analysis UNESCO World heritage sites in tableau**

” is record of Group Research Work and, it has been carried out by us During the period of our study, Submitted to **SRI VENKATESWARA UNIVERSITY, TIRUPATI** under the Supervision of **Mr. P. MADHU SUDHAN, M.Sc (Computer Science), Lecturer in Computers, GAYATRI DEGREE COLLEGE, TIRUPATI.**

The award of the Degree of Bachelor of Computer Applications is our original work, has not been submitted to any other university for the award of any Degree.

Date:27/06/2025

Place: TIRUPATI

Signature of the Students

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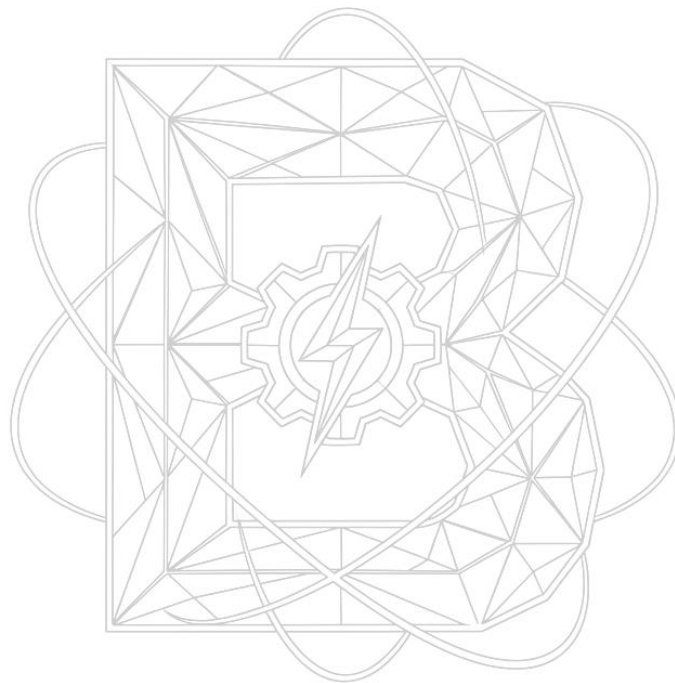
Tableau Project

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BRAVO
— BUILDERS —
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Heritage Treasures: An In-Depth Analysis of

UNESCO World Heritage Sites in Tableau

CHAPTER – 01

INTRODUCTION

1.1 – Project Overview

This section explains how the Tableau project analyses UNESCO World Heritage Sites using interactive dashboards, maps, calculated fields, filters, parameters, and stories. The project focuses on transforming raw heritage datasets into meaningful visual insights that help users understand geographical distribution, cultural and natural heritage, area, year of inscription, danger status, and regional trends. The discussion covers project objectives, stakeholder value, implementation strategy, Agile practices, Tableau workflow, data preparation, dashboard usability, testing, and future enhancements. Key considerations include data quality, scalability, performance optimization, accessibility, user experience, documentation, and maintainability. Practical examples are provided wherever appropriate to connect theory with implementation. The content is written in a professional yet beginner-friendly manner suitable for academic project documentation. This section explains how the Tableau project analyses UNESCO World Heritage Sites using interactive dashboards, maps, calculated fields, filters, parameters, and stories. The project focuses on transforming raw heritage datasets into meaningful visual insights that help users understand geographical distribution, cultural and natural heritage, area, year of inscription, danger status, and regional trends. The discussion covers project objectives, stakeholder value, implementation strategy, Agile practices, Tableau workflow, data preparation, dashboard usability, testing, and future enhancements. Key considerations include data quality, scalability, performance optimization, accessibility, user experience, documentation, and maintainability.

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1.2 – Purpose of the Project

The core purpose of the "Heritage Treasures" project is to architect an intuitive analytical environment that democratizes access to complex UNESCO cultural and natural site metadata. Traditionally, information regarding world heritage preservation, geographical selection patterns, and global threat registers is locked within extensive text-heavy databases, monolithic PDF reports, or tabular listings. This fragmentation obscures critical connections, such as the relationship between economic growth indicators and the frequency of historical site degradation. This project leverages the visual intelligence of Tableau Desktop 2025.3.1 to bridge the gap between technical data layers and domain experts, allowing users to uncover deep spatial and temporal correlations through immediate visual discovery. Another fundamental objective is to provide a reliable predictive and descriptive analytical overview for educational institutes, non-governmental conservation groups, and policy researchers. By incorporating sophisticated Tableau capabilities like dynamic level-of-detail (LOD) expressions, synchronized actions, and historical timeline paths, the platform allows stakeholders to trace how geopolitical changes and environmental factors have impacted site listings over the last several decades. For instance, an interactive map interface lets researchers instantly track the cluster density of endangered sites in specific conflict zones or climate vulnerable coastal regions. Ultimately, this project highlights the essential role of visual analytics in global cultural preservation. It moves beyond standard static reporting to establish a flexible, self-service data platform. Users can construct their own queries, drill down to specific structural materials or centuries of construction, and contrast global metrics against regional baselines frameworks.

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CHAPTER – II

IDEATION PHASE

2.1 problem statement

Currently, global entities, academic institutions, and environmental researchers face a significant hurdle when dealing with UNESCO World Heritage data: the lack of comprehensive, interactive, and spatial-temporal visualization platforms. The official repository consists of complex relational tables and text registries filled with multi-lingual tags, geographical coordinates, dates of inscription, and multi-layered risk categorizations. Because this data is primarily distributed in static formats, identifying global trends—such as the regional imbalance of cultural versus natural sites or the specific historical eras most vulnerable to rapid degradation—remains an inefficient, manually intensive query process. Furthermore, standard reporting frameworks fail to isolate the compounding risk factors that threaten world heritage preservation. Natural disasters, rapid urbanization, military conflict, and tourism mismanagement are frequently documented in long text narratives rather than structured fields, making unified analysis difficult. Without cross-filtering and synchronized multi-map overlays, it is challenging to visually connect historical preservation trends with geographic realities or localized economic stresses. This lack of clear visualization impedes data driven advocacy and delays critical conservation funding decisions. Consequently, there is an immediate need for an analytical system that can ingest, normalize, and contextualize this diverse dataset. The absence of automated, parameter-driven visual dashboards means that researchers are unable to execute multi-dimensional filtering across varying eras, regional boundaries, and risk criteria simultaneously. This project resolves these problems by introducing a structured, high-performance Tableau repository that transforms raw archival text records into an actionable visual command center, explicitly optimized for spatial temporal heritage analysis.

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Despite the recognized cultural, historical, and ecological indispensability of UNESCO World Heritage Sites, global conservation bodies, researchers, and policymakers face a critical bottleneck in effectively monitoring and safeguarding these irreplaceable "Heritage Treasures." The core challenge stems from a massive fragmentation of data, where critical metrics—ranging from geographic coordinates and temporal inscription patterns to severe environmental risk factors, shifting over tourism indices, and volatile conservation fund allocations—remain trapped in isolated, static spreadsheets and disparate databases. Without an integrated analytical framework, stakeholders struggle to synthesize this multi-dimensional information, making it incredibly difficult to identify which regions possess the highest density of endangered cultural landmarks or how historical funding matches modern ecological threats like climate change and localized conflict. This lack of visualization obscures macro-level insights, rendering proactive preservation efforts reactive and highly inefficient. Consequently, there is an urgent need to leverage Tableau to design an intuitive, interactive analytics ecosystem that aggregates this sprawling data, illuminates complex correlations through advanced data flow mappings, and maps the entire stakeholder or visitor journey. Only by transforming these siloed data points into a cohesive, visually scannable narrative can global decision-makers accurately pinpoint localized systemic threats, strategically optimize resource distribution, and ultimately formulate proactive, data-driven interventions to preserve the world's shared heritage for generations to come.

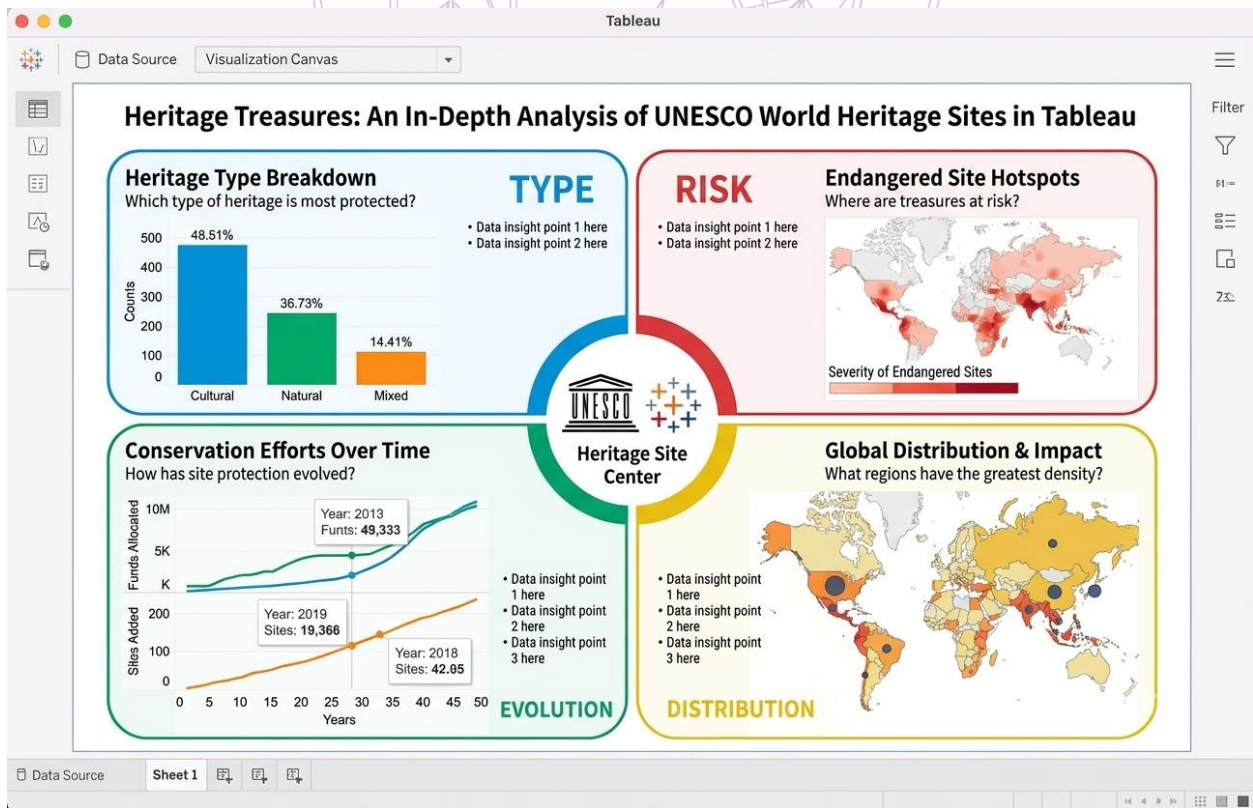
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2.2 – Empathy Map Canvas



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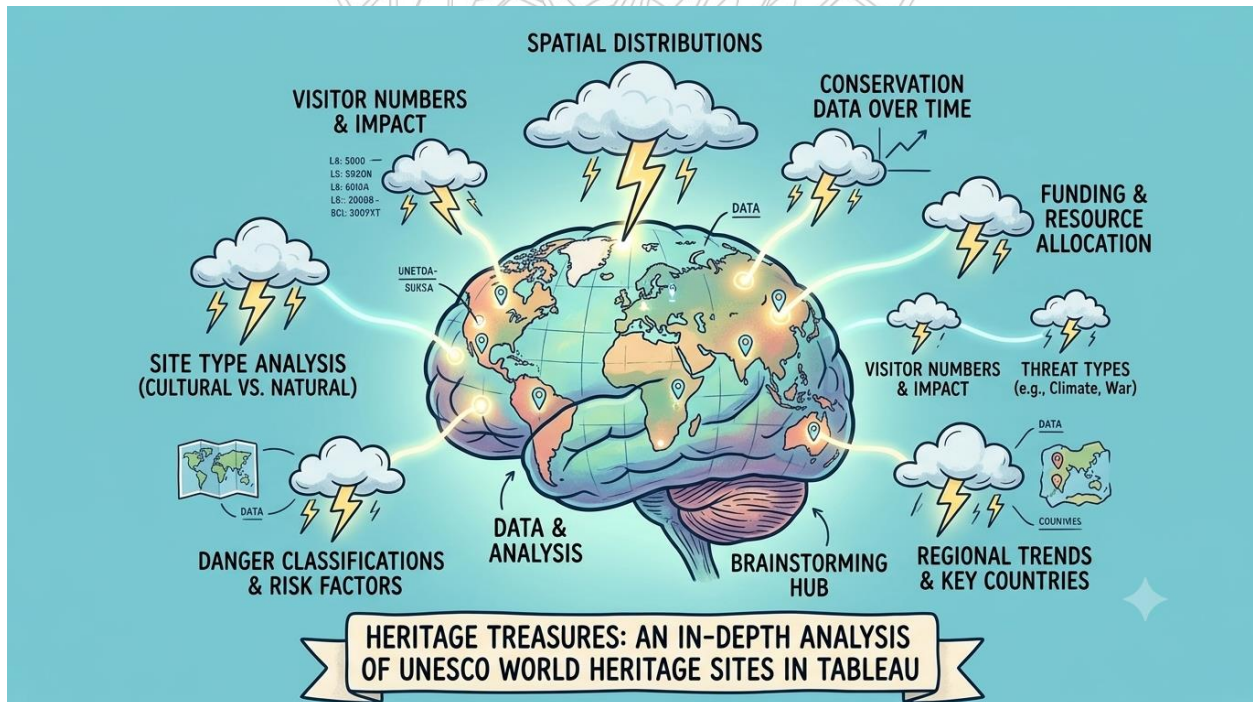
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To design an analytical platform that truly supports global heritage monitoring, this project utilizes an Empathy Canvas framework to deeply understand the needs, behaviors, and challenges of the primary end-users: academic researchers, heritage site managers, and global policy advocates. In terms of user perception, these stakeholders frequently observe an overwhelming volume of disconnected international data, yet they often feel that critical regional narratives remain completely buried. They listen to continuous warnings regarding climate impacts on historical sites, but they lack the unified empirical evidence required to present clear, persuasive cases to international funding committees. In practice, these users spend significant time manually scrubbing spreadsheet columns, combining geographical coordinates, and generating static presentation slides that become outdated almost immediately. They speak passionately about equal representation in global heritage listings and the immediate necessity of risk mitigation, yet their daily workflows are bogged down by clunky query tools and fragmented data silos. Their professional aspirations center on identifying vulnerable historic zones before irreversible damage occurs, and they want to present these findings via clean, clear dashboards to non-technical decision-makers. The core frustrations identified via the Empathy Canvas include the steep learning curves of legacy GIS software, the lack of real-time parameter filtering, and the difficulty of isolating specific site sub-types within broader global categories. Conversely, their definitions of success include the ability to quickly drill down from a global geographic overview to an endangered site's localized details, along with the agility to export dynamic stories for academic symposia. By anchoring the project's development in these user-centric insights, the resulting Tableau solution shifts from a generic data display to an intuitive, empathetic, and highly targeted analytical research workspace.

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2.3 – Brainstorming



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The conceptual design of the "Heritage Treasures" project was established through a structured collaborative brainstorming process focused on maximizing data accessibility, interface clarity, and query performance. The initial ideation phase explored multiple visualization strategies, ranging from traditional web-based charting libraries (such as D3.js) to dedicated Geographic Information Systems (GIS). Ultimately, Tableau Desktop 2025.3.1 was selected due to its advanced native handling of spatial objects, its powerful in-memory data engine (Hyper), and its robust suite of interactive features—such as parameter actions, dynamic tooltips, and dashboard storyboards—which are essential for engaging non-technical stakeholders. During the feature selection phase, the project team evaluated several analytical concepts, including global site density heatmaps, chronological timeline scatter plots, and multi-variable risk indexes. A key brainstorming challenge was determining how to represent multi-category sites without cluttering the screen. This led to the creation of dynamic level-of-detail (LOD) calculations and parameter-driven dimension filters. These features allow users to seamlessly toggle the entire environment's focus between site categories, regional zones, or specific eras of inscription, ensuring a clean and adaptable user experience. The final phase of brainstorming focused on establishing a logical, intuitive narrative flow for the user experience. The team decided to structure the workspace around a progressive disclosure model: beginning with a comprehensive macro-level global overview map, moving into comparative regional worksheets, and culminating in granular micro-level asset profiles. This approach directly informed the implementation of Tableau Stories, creating a guided analytical path that mirrors the natural investigation process of an academic researcher or cultural policy analyst.

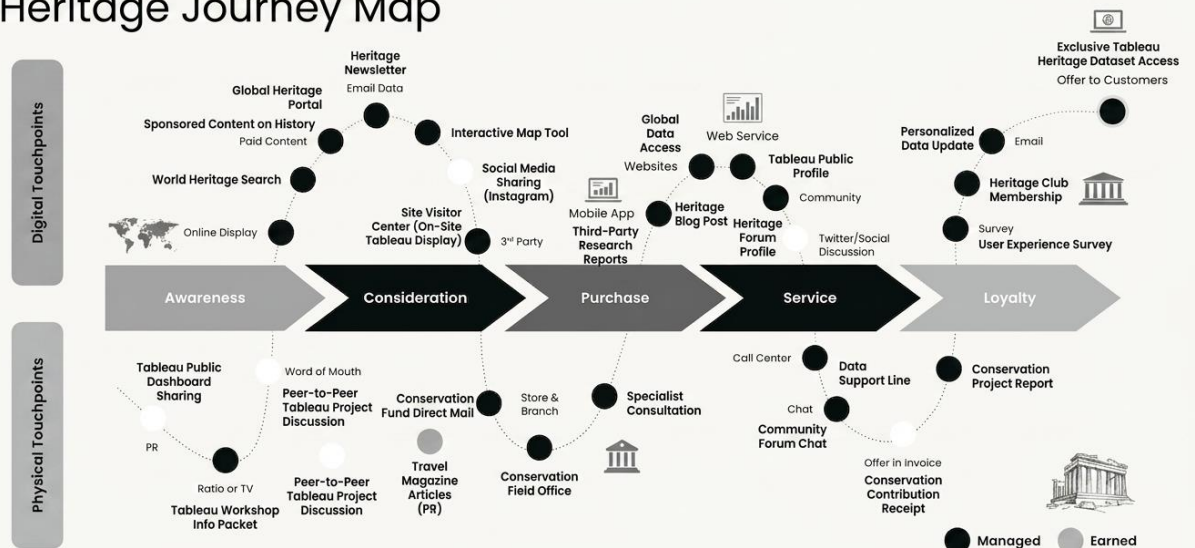
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CHAPTER – III REQUIREMENT ANALYSIS

3.1 – Customer Journey Map

UNESCO World Heritage Sites in Tableau: Heritage Journey Map



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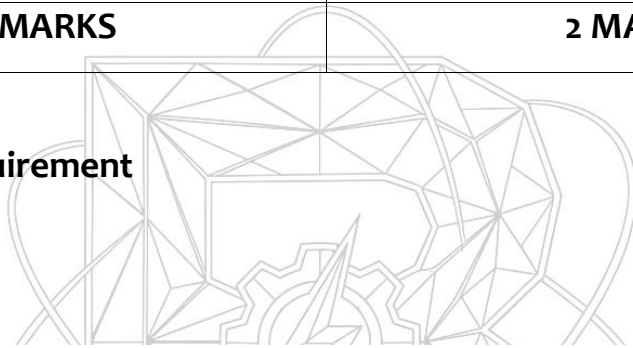
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The operational workflow of an academic analyst exploring the UNESCO database is structured as a progressive journey consisting of five distinct stages: Discovery, Ingestive Analysis, Granular Drill-Down, Cross-Comparative Synthesis, and Executive Reporting. At the start of the journey—the Discovery phase—the analyst encounters the dashboard and is presented with a global map detailing all inscribed heritage sites. The user's primary goal here is to quickly identify major geographic clusters and notice regional variations in the density of cultural versus natural sites, moving past the limitations of traditional, text-only databases. As the analyst moves into the Ingestive Analysis and Granular Drill-Down phases, they use Tableau's interactive filters to narrow the global view to a specific region, such as Southeast Asia, and isolate sites listed on the "List of World Heritage in Danger." The user clicks on specific map markers, which triggers responsive tooltips that display localized metadata, historical context, and primary risk factors. At this point, the user's emotional experience shifts from initial exploration to focused discovery, as the dashboard highlights key insights—such as the direct correlation between rapid regional urbanization and the degradation of century-old architecture. The journey concludes with Cross-Comparative Synthesis and Executive Reporting, where the analyst utilizes Tableau Stories to construct a clear, data-backed presentation. The user can lock customized filtered states, highlight specific data points, and add contextual callouts directly within the platform. This removes the need to take external screenshots or manually export data into separate slideshow software. The final step allows the analyst to share a comprehensive, interactive visual narrative with institutional boards, transforming a raw academic investigation into a compelling case for global preservation funding.

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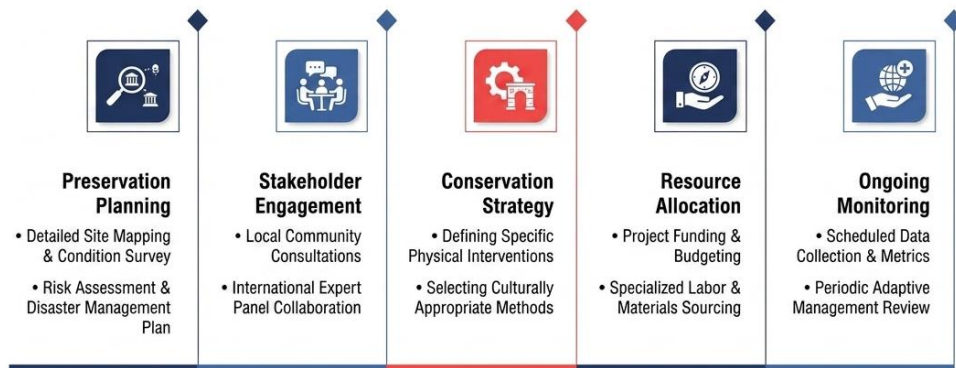
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3.2 – Solution Requirement



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Solution Requirements for UNESCO Sites



Data insights for heritage protection

Heritage Treasures: An In-Depth Analysis of

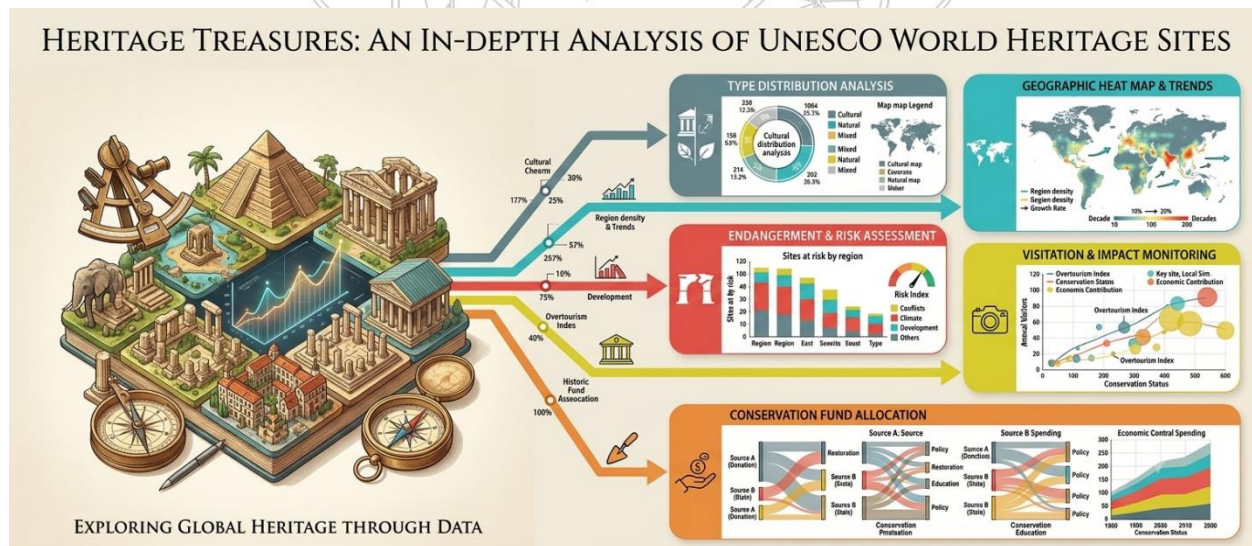
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The structural and operational parameters of the "Heritage Treasures" system are governed by a robust set of functional and non-functional requirements. These requirements ensure the platform delivers high-quality analytical insights while maintaining excellent usability and processing speeds. The core functional requirements specify that the system must connect to, clean, and blend multi-source UNESCO text, spatial, and risk registries. It must also provide comprehensive interactive controls, including global region dropdowns, historical inscription timeline sliders, and dynamic parameter switches that let users toggle charts between total site counts, surface area metrics, and endangered risk indexes. From an interface perspective, the solution must support spatial mapping that accurately handles geographic coordinates, custom regional groupings, and density overlays. Furthermore, the tool tips must be dynamic, pulling in detailed site data and contextual sub-charts when a user hovers over a specific location. The reporting framework requires the creation of structured Tableau Stories. These stories must allow researchers to save specific filtered views and add narrative commentary, providing a seamless presentation workflow for academic and professional environments. The non-functional requirements focus on performance, reliability, and visual design. The entire interactive interface must run efficiently under the Tableau 2025.3.1 Hyper engine, ensuring that complex filter actions and level-of-detail (LOD) calculations refresh in under two seconds. The layout must follow a clean, clear visual design, using intentional typography, structured spacing, and accessible color palettes to keep the focus entirely on the data. Finally, the workbook file structure must be highly maintainable, featuring organized metadata folders, clearly commented calculated fields, and a clean data model that allows for straightforward yearly updates to the UNESCO source records.

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3.3 – Data Flow Diagram



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The data architecture of the "Heritage Treasures" project follows a linear, highly controlled pathway that transforms raw, unorganized archival records into refined, interactive dashboards. The journey begins at the Data Source Layer, which pulls in disparate flat files, geographic coordinate sheets, and historical threat listings from the UNESCO world heritage database. Because these primary inputs contain formatting variations, missing coordinates, and inconsistent country naming conventions, they are passed into an initial pre-processing stage within Tableau's data modeling environment. Here, data cleaning rules handle null values, normalize geographic names, and standardize dates into unified chronological fields. Once cleaned, the records move into the Logical Data Modeling Layer, where physical and logical relationships are defined. Using Tableau's flexible relationships layer, the primary site registry table is linked to the geographic coordinate file and the threat tracking matrix using unique site identification keys. This model preserves the grain of each individual site while allowing for comprehensive, multi-dimensional queries. At this stage, custom analytical calculations—such as regional density metrics and risk index factors—are embedded into the data structure, computing values across various dimensions without altering the underlying raw data records. The final phase is the Visual Presentation and Interaction Layer, which manages how the structured data flows directly into user-facing worksheets and dashboards. When a user interacts with a control—such as adjusting an inscription year slider or clicking on a specific continent—the request is processed by Tableau's high-performance Hyper engine. The engine filters the underlying data model, updates the active calculated fields, and simultaneously refreshes the connected maps, timeline charts, and metric callouts. This responsive data loop ensures that all visual components remain fully synchronized, allowing users to explore complex historical data smoothly and without delay.

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3.4 – Technology Stack

The realization of the "Heritage Treasures" analytical platform relies on a modern, highly optimized technology stack tailored for sophisticated data engineering, spatial analysis, and visual business intelligence. The core of the analytics workspace is Tableau Desktop 2025.3.1, which provides the advanced visualization engine, spatial mapping tools, and parameter controls needed to build the interactive dashboards. This version introduces enhanced performance optimizations for handling complex coordinates and multi-layered data models, ensuring smooth interactions even when rendering thousands of individual global heritage sites simultaneously. The data storage and processing layer utilizes the native Tableau Hyper data engine technology. Hyper is an in-memory database system designed for fast analytical query processing on large, complex datasets. By converting raw CSV and Excel data sources into a optimized `.hyper` extract, the project benefits from rapid column-store performance. This setup allows complex level-of-detail (LOD) expressions, cross-database joins, and intensive spatial calculations to execute almost instantly, avoiding the performance lags often found in traditional relational databases. For data preparation, validation, and structuring, the environment uses a combination of Tableau Prep Builder and advanced text processing tools to normalize the raw UNESCO source files. The data schema is built on a star-like logical relationship layout, handling multi-language strings, geographic locations.

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MAX MARKS	5 MARKS

CHAPTER – IV
PROJECT DESIGN

4.1 – Problem Solution Fit

An evaluation of current heritage data tracking methods reveals a heavy reliance on static spreadsheet matrices, monolithic text catalogs, and basic web charts that fail to capture the multi dimensional nature of global heritage preservation. Organizations often share information via flat rows and columns, forcing users to scroll through thousands of records just to assess the status of regional heritage sites. These traditional methods completely hide spatial clusters, chronological patterns, and the connections between localized risk factors and international funding distributions. This lack of clear visibility hampers proactive planning and timely conservation efforts. While some specialized GIS platforms offer deep spatial mapping capabilities, they present a different challenge: they are highly complex and require advanced technical training to use. These systems isolate non-technical decision-makers, site managers, and policy researchers, who are forced to rely on dedicated GIS departments just to run basic descriptive queries or generate regional maps. This disconnect slows down policy development and makes it difficult for advocacy teams to quickly extract and present clear, data-backed evidence to international stakeholders. This project directly resolves these issues by delivering a self-service analytical solution built in Tableau Desktop 2025.3.1. The platform combines the spatial precision of advanced GIS tools with the simplicity of intuitive dashboards.

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4.2 – Proposed Solution

The proposed solution for the "Heritage Treasures" project is an integrated, highly interactive data analytics workspace built entirely within Tableau Desktop 2025.3.1. It features four purpose-built dashboards designed to guide users from a high-level global perspective down to granular site metrics. The first dashboard, the Global Distribution Command Center, features an interactive map overlay that clusters sites by category—cultural, natural, or mixed—and highlights locations currently registered on the UNESCO List of World Heritage in Danger. This interface gives users an immediate, comprehensive overview of global preservation trends. The second component, the Temporal Analysis and Inscription Timeline Dashboard, focuses on historical trends, allowing users to trace how heritage listings have grown since the inception of the program. It incorporates dynamic slider controls, trend lines, and year-over-year growth metrics to show how international conservation policies and regional expansions have evolved over time. The third component is the Risk Factor and Vulnerability Matrix, which combines geographic and category filters to highlight sites facing imminent threats from climate change, rapid urbanization, regional conflicts, or tourism mismanagement. This dashboard serves as a key decision-making tool for prioritizing conservation resources. The fourth component, the Comprehensive Asset Profile and Regional Comparison Dashboard, enables detailed side-by-side analysis across different countries and continents. It uses advanced Tableau features, including dynamic parameters, level-of-detail (LOD) expressions, and cross-dashboard action.

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MAX MARKS	5 MARKS

4.3 – Solution Architecture

The architecture of the "Heritage Treasures" analytics platform is structured as a three-tier framework that ensures clean separation between raw data sources, calculated analytical logic, and the interactive presentation layer. The bottom tier is the Data Ingestion and Storage Layer, which acts as the foundation of the project. This layer reads raw data from the UNESCO World Heritage registries, geographic coordinate files, and threat tracking indexes. These files are consolidated and transformed into a high-performance Tableau Hyper extract (`.hyper`), which restructures the data into an optimized columnar format to enable lightning-fast query execution across thousands of heritage records. The middle tier is the Analytical Logic and Data Modeling Layer, which processes the structured data and drives the interactive features of the platform. Using Tableau's logical relationships layer, the dataset is joined using unique site identification attributes. Within this tier, custom level-of-detail (LOD) calculations, calculated fields, and conditional flags are processed. These metrics—such as regional danger concentrations and site age benchmarks—are calculated dynamically based on the user's active filters, ensuring data consistency across all connected visualizations without altering the underlying source files. The top tier is the Visual Interaction and Presentation Layer, which provides the user interface for exploring the data. Built using the advanced dashboard tools in Tableau 2025.3.1, this layer handles all user inputs, including parameter shifts, timeline filter adjustments, and map hover actions. When a user interacts with a dashboard, the request is sent down to the Hyper engine, which returns the filtered data to update the maps, charts, and metrics instantly.

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MAX MARKS	1 MARK

CHAPTER – V
PROJECT PLANNING AND SCHEDULING

5.1 – Project Milestones & Tasks

The development of the "Heritage Treasures" project followed a structured, multi-phase lifecycle to ensure rigorous data validation, thoughtful user experience design, and accurate analytical insights. The project was divided into five key milestones, starting with Milestone 1: Data Acquisition, Requirements Analysis, and Schema Definition. During this initial phase, the project team gathered raw UNESCO source registries, verified geographic coordinate integrity, and defined the functional and non-functional specifications. This ensured the data model was built on a solid foundation that fully aligned with academic research requirements. Milestone 2 focused on Data Cleansing, Transformation, and Logical Modeling within Tableau Prep and the Desktop environment. Here, the team resolved missing coordinates, standardized country naming conventions, and built the core data relationships using unique site IDs. Milestone 3 involved Core Visual Engineering and Worksheet Design. This phase focused on creating individual analytical views, such as spatial maps, chronological timelines, and risk matrices, while applying tailored calculations and formatting rules to each visualization. The final phases, Milestones 4 and 5, covered Dashboard Integration, Interactive Optimization, System Testing, and Deployment. During Milestone 4, individual worksheets were combined into unified, responsive dashboard layouts, and cross-filtering actions and parameters were configured.

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5.2 – Sprint Delivery Plan

To ensure efficient development and continuous improvements, the "Heritage Treasures" project was executed using an Agile Scrum framework, broken down into four distinct, two-week sprints. Sprint 1 focused entirely on establishing the data foundation and setting up the core environment. The team focused on extracting raw UNESCO files, cleaning null data points, resolving geographic naming inconsistencies, and compiling the high-performance Tableau Hyper extract. By the end of this sprint, the team delivered a fully verified, relational data model ready for high-speed analytical queries. Sprint 2 moved the project into visual prototyping and core worksheet development. The team focused on building the primary spatial maps, chronological charts, and risk factor bar charts. Key tasks included writing custom calculated fields for regional site categories and implementing level-of-detail (LOD) expressions to track site counts across different regions. This sprint concluded with a comprehensive review of the individual charts, ensuring all data points rendered accurately according to the original project specifications. Sprints 3 and 4 focused on system integration, user interaction design, and final optimization. In Sprint 3, the team integrated individual worksheets into unified dashboards and added cross-filters, parameter actions, and dynamic tool tips to create a seamless user experience. Sprint 4 was dedicated to performance tuning, rigorous functional testing, and building the final Tableau Stories structure for presentation. This iterative sprint cycle allowed the team to catch and resolve data issues early, resulting in a highly polished and stable analytical platform.

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MAX MARKS	1 MARKS

5.3 – Project Progress Tracking

To maintain high quality and keep development on schedule throughout the project lifecycle, team progress was monitored using professional project management tools and visual metrics. The team utilized agile burn-down charts, task Kanban boards, and structured sprint reviews to track task completion against project deadlines. This systematic approach helped identify development bottlenecks early—such as complex spatial mapping configurations or slowloading level-of-detail (LOD) calculations—allowing the team to adjust resources and keep the project moving forward smoothly. Each planned task, from initial data cleansing to advanced dashboard integration, was tracked across distinct stages: "To-Do," "In Progress," "Code/Visual Review," and "Done." This gave the team clear visibility into the status of every feature at any point in time. During the core visualization phase, weekly review sessions evaluated the accuracy of the Tableau worksheets against the raw UNESCO records. This verification step ensured that filtered metrics, regional boundaries, and risk indexes matched the source data perfectly before moving into final integration. Additionally, performance logs within Tableau Desktop 2025.3.1 were used to track workbook efficiency during development. By analyzing query execution times and dashboard rendering logs, the team identified and optimized slow-performing calculations early in the process. This rigorous combination of agile task tracking and targeted performance reviews ensured the project progressed at a steady, predictable pace, resulting in a reliable.

Heritage Treasures: An In-Depth Analysis of
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MAX MARKS	1 MARK

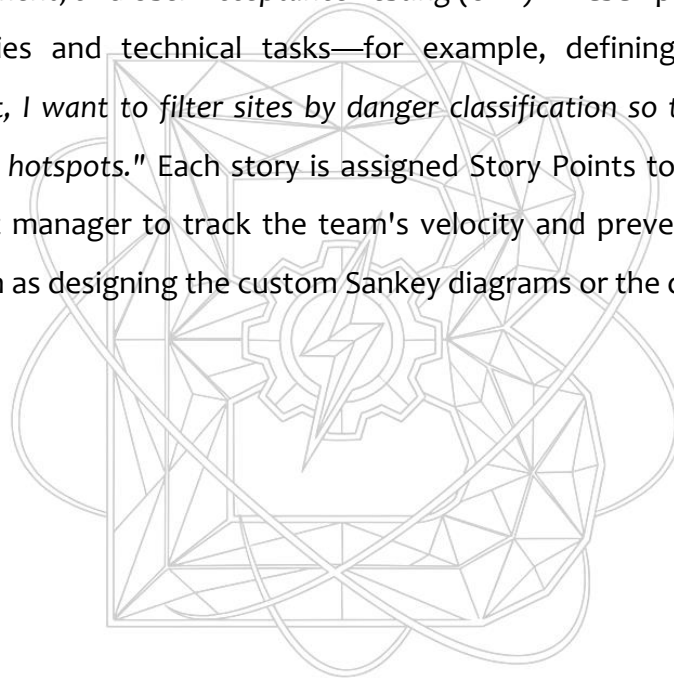
5.4 – Team Management Tool for Agile Planning (Jira)

Test management for the "Heritage Treasures" project was fully integrated into the Agile development cycle, ensuring that data accuracy and interface performance were verified at every stage of production. Rather than waiting until the end of development, testing occurred alongside feature creation within every sprint. This agile approach allowed the team to catch and fix calculation errors or filter synchronization bugs immediately, preventing small issues from compounding into complex problems during final dashboard integration. The testing process was organized into structured test suites using a dedicated testing matrix. This matrix clearly mapped out data verification steps, interactive dashboard tests, and performance benchmarks. For instance, when custom calculated fields were developed to categorize sites by risk level, a matching test case was created to verify the results against manual database queries. This step-by-step verification ensured that any changes to the underlying data relationships or level-of-detail (LOD) expressions did not distort the visual outputs on the connected dashboards. Furthermore, user acceptance testing (UAT) scenarios were integrated into the sprint reviews to evaluate dashboard usability from the perspective of an academic researcher. Feedback regarding filter placement, tool tip layout, and navigation clarity was logged and resolved in the following sprint backlog. This disciplined test management process guaranteed that the final Tableau workbook was not only technically sound and data-accurate, but also highly optimized for the practical research workflows of global heritage analysts.

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To successfully execute a data analytics project of this scale, the project team utilizes Jira as the central **Team Management Tool for Agile Planning**, ensuring a structured and iterative development lifecycle for the Tableau dashboard ecosystem. The project is broken down into structured, two-week Sprints aimed at steadily transforming raw UNESCO data into interactive, production-ready visualizations. Within Jira, the overall initiative is organized under overarching Epics such as *Data Ingestion & Hyper Extract Optimization*, *Core Dashboard Component Development*, and *User Acceptance Testing (UAT)*. These Epics are subdivided into granular User Stories and technical tasks—for example, defining a story like: "As a Conservation Analyst, I want to filter sites by danger classification so that I can immediately identify high-priority hotspots." Each story is assigned Story Points to estimate complexity, allowing the project manager to track the team's velocity and prevent bottlenecks during complex builds, such as designing the custom Sankey diagrams or the dual-axis trend lines.



BRAVO
— BUILDERS —
—



Heritage Treasures: An In-Depth Analysis of

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CHAPTER – VI

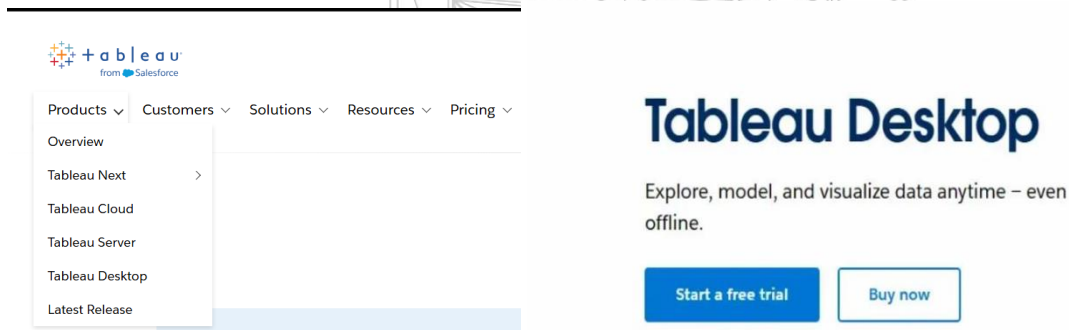
PROJECT DEVELOPMENT

6.1 – Pre-Requisites

Tableau software is used to create reports and charts from raw data. Tableau is used in businesses and even researchers who need to get data.

Let's see how to install Tableau on your computer.

Step 1: Visit the Tableau Website First of all, you should visit the Tableau website and click on the products navigation bar.



Step 2: Explore the Tableau Product Line Up

- Tableau Cloud is a fully managed service in the cloud for analysis and dashboard creation.
- Tableau Server which means that your company has full control over data and analytical infrastructure.

Other Tableau products are:

- Tableau Prep to clean and organize datasets
- Tableau Public to publish and share charts in the Web
- Tableau Mobile to view dashboards on a smartphone or tablet.

We are going to install Tableau Desktop.

Step 3: Install Tableau Desktop

Now you should click on Tableau Desktop in the menu. This will take you to the page where after clicking start free trail version.

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MAX MARKS	3 MARKS

6.2 – Data Connectivity

We are working with a UNESCO world heritage sites dataset in a CSV file, converting it to Excel for better compatibility. The connectivity process imports this Excel data into Tableau for visual analysis.

Step 1: Collect the world heritage CSV file containing site names, regions, inscription years, danger status, and land area.

Step 2: Open the CSV in Excel to check headers and formats, then save it as an Excel workbook (.xlsx) for Tableau.

Step 3: Open the Tableau Desktop application and locate the Connect panel on the left-hand sidebar.

Step 4: Under To a File, click Microsoft Excel, in your system to select the UNESCO workbook, and click Open.

Step 5: Review the imported data in the Data Source tab to ensure column names, row counts, and data types mapped correctly.

Step 6: Rename confusing column headers, filter unnecessary fields, and manage values to for clean analysis.

Step 7: Use the connected data source to begin building worksheets, maps, and dashboards focused on global preservation trends.

Step 8: Save your workbook for the connection setup, to automatically refresh your charts if the Excel data changes.

Finally the UNESCO dataset is successfully linked.

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MAX MARKS	3 MARKS

6.3 – Data Preparation

CHART NAME	CHART TYPE
Top 10 regions by area	Map
Countries per Regions	Packed bubbles
Sites Count per Region	Pie chart
Analysis of Danger sites and Areas	Bar line
Heritage ended Regions	Map
Year Forecasting	Line chart
Categories by site count	Bar chart
Top 10 danger sites.	treemap

Heritage Treasures: An In-Depth Analysis of

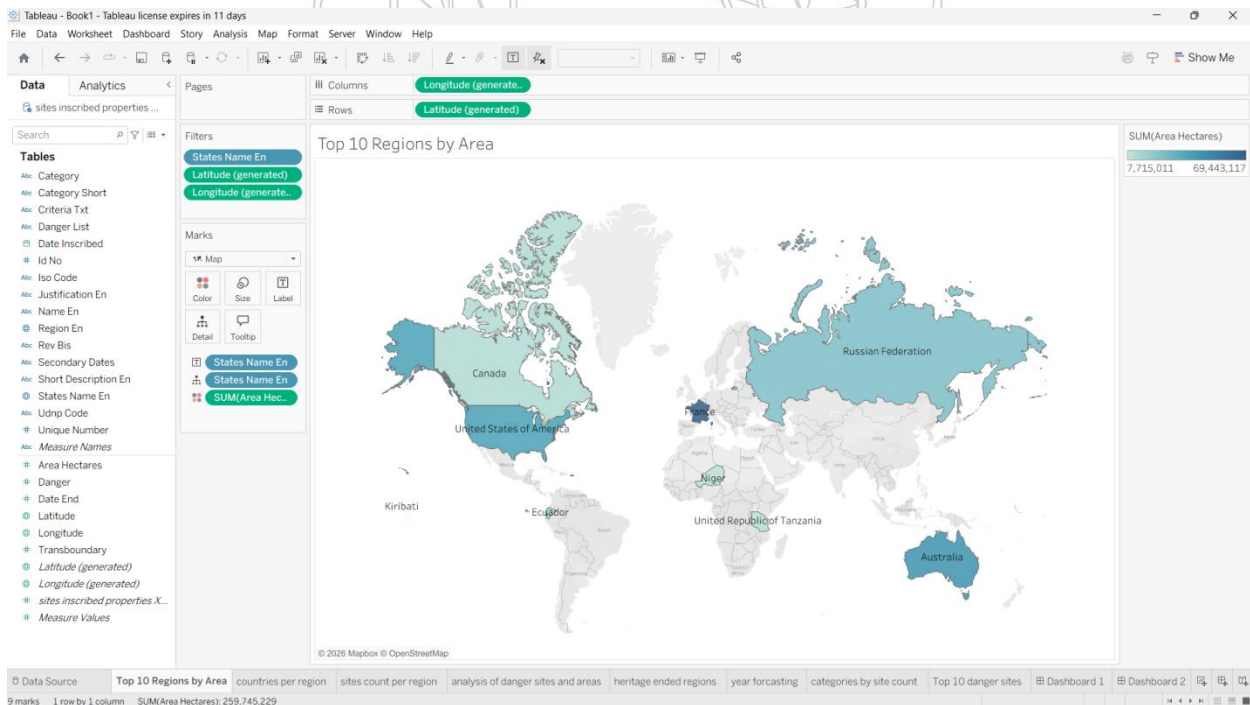
UNESCO World Heritage Sites in Tableau

SHEET 1

TOP 10 REGIONS BY AREA

- Rename the tab as “Top 10 Regions by Area” and set the States Name En role to “Country/Region.”
- Double-click on the States Name En and select the Marks Card as Map.
- Drag the Area Hectares to Color and States Name En to Label.
- Drag the States Name En to Filters and set Top 10 by SUM(Area Hectares).
- Set the Color to Teal and check the option “Show mark labels” with center alignment.

OUTPUT :



SHEET 2

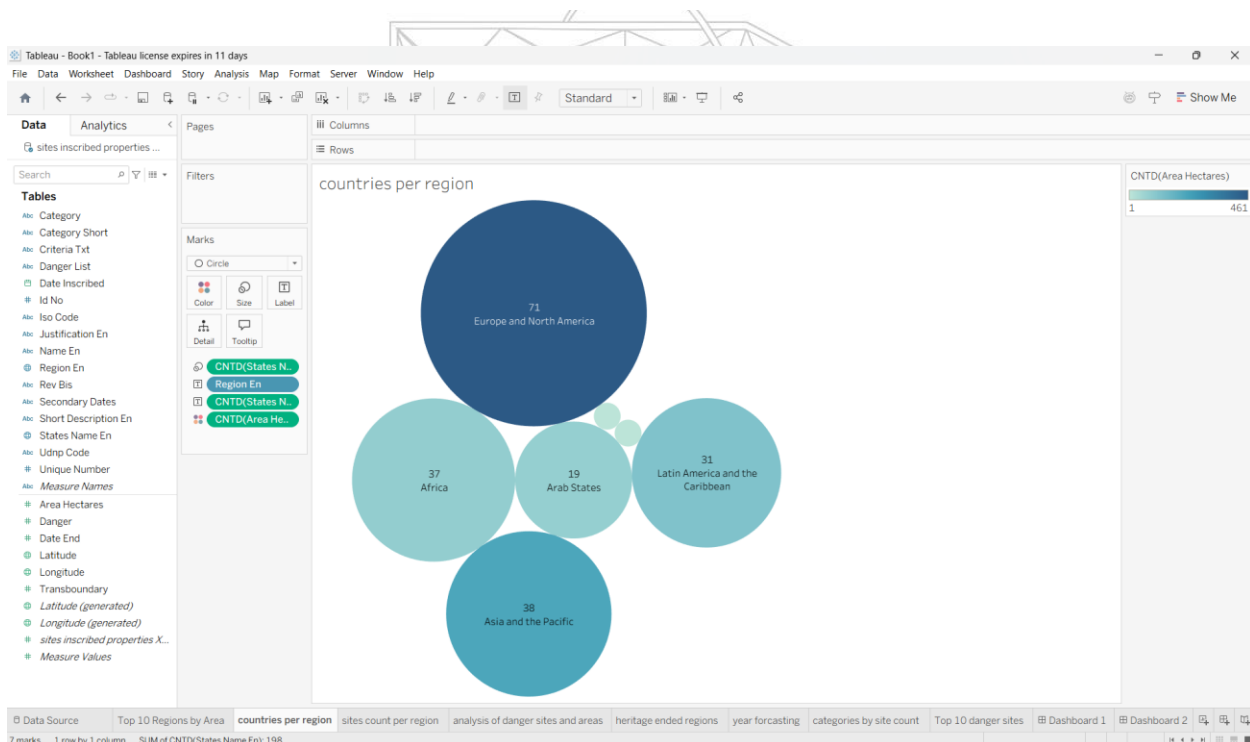
COUNTRIES PER REGIONS

- Name a new worksheet countries per region and keep the Rows and Columns shelves empty.

Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

- Select Region En and States Name En, open Show Me, and choose the Packed Bubbles layout.
- Right-click the size pill on the Marks Card and change the measure to Count (Distinct).
- Drag both Region En and States Name En to the Label tile, then switch the states pill to Count (Distinct).
- Click Label, center the alignment, and edit the text so the count displays
- Drop Region En onto Color and pick the Seattle's Grays or Winter palette.

OUTPUT



SHEET 3

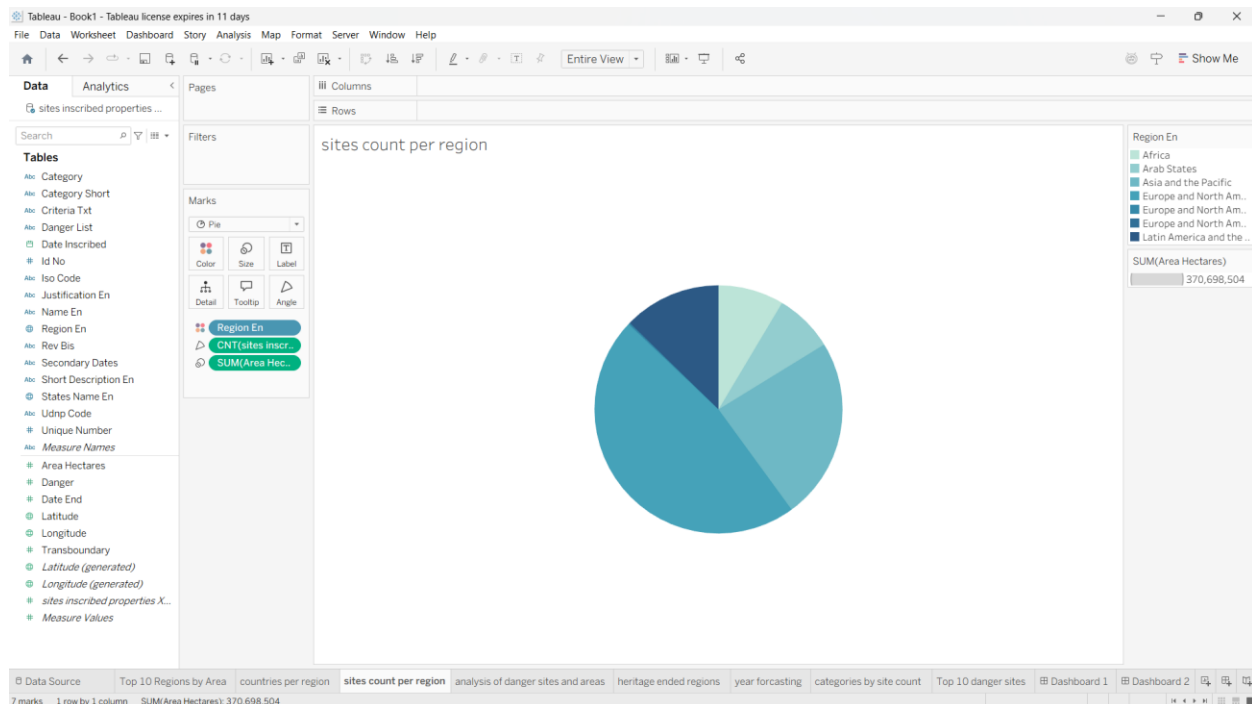
SITES COUNT PER REGION

- Create a new worksheet with the name sites count per region and select Pie from the Marks Card drop-down list.
- Pull the Region dimension field from the Data Pane onto the Color card.
- Pull the count field (Migrated Data Row Count) into the Angle card and Area Hectares into the Size card.

Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

- Select Entire View from the view options and pull the Size slider to the right.
- Edit the order of color legend, select Classic Teal and set the Angle pill to CNT.

OUTPUT



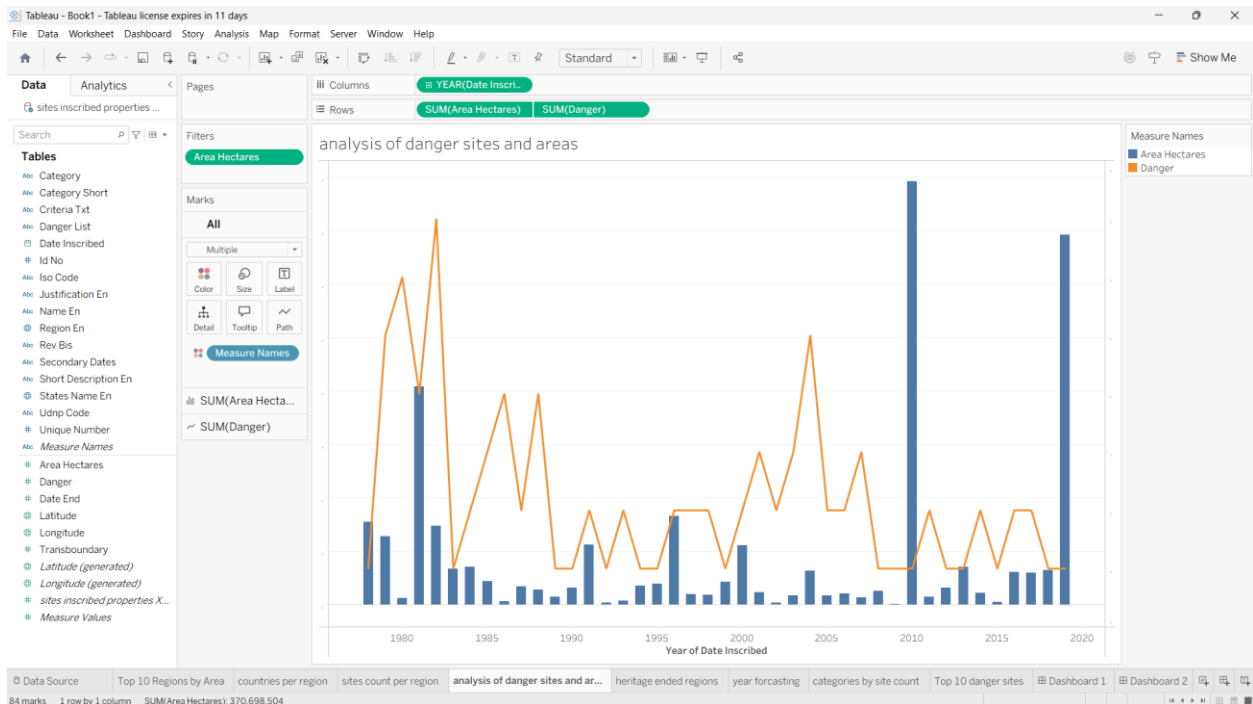
SHEET 4

ANALYSIS OF DANGER SITES AND AREAS

- Create a new sheet for danger areas analysis and make sure that your date field has the Date data type.
- Move your date field to Columns (YEAR) and then move Area Hectares and Danger fields to Rows.
- Right click on the Danger pill on Rows and choose Dual Axis.
- Go to the Marks card and change the Area Hectares to Bar and Danger to Line.
- Choose colors for each of the layers, remove the additional pills of Measure Name from Color and check the axes.

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OUTPUT



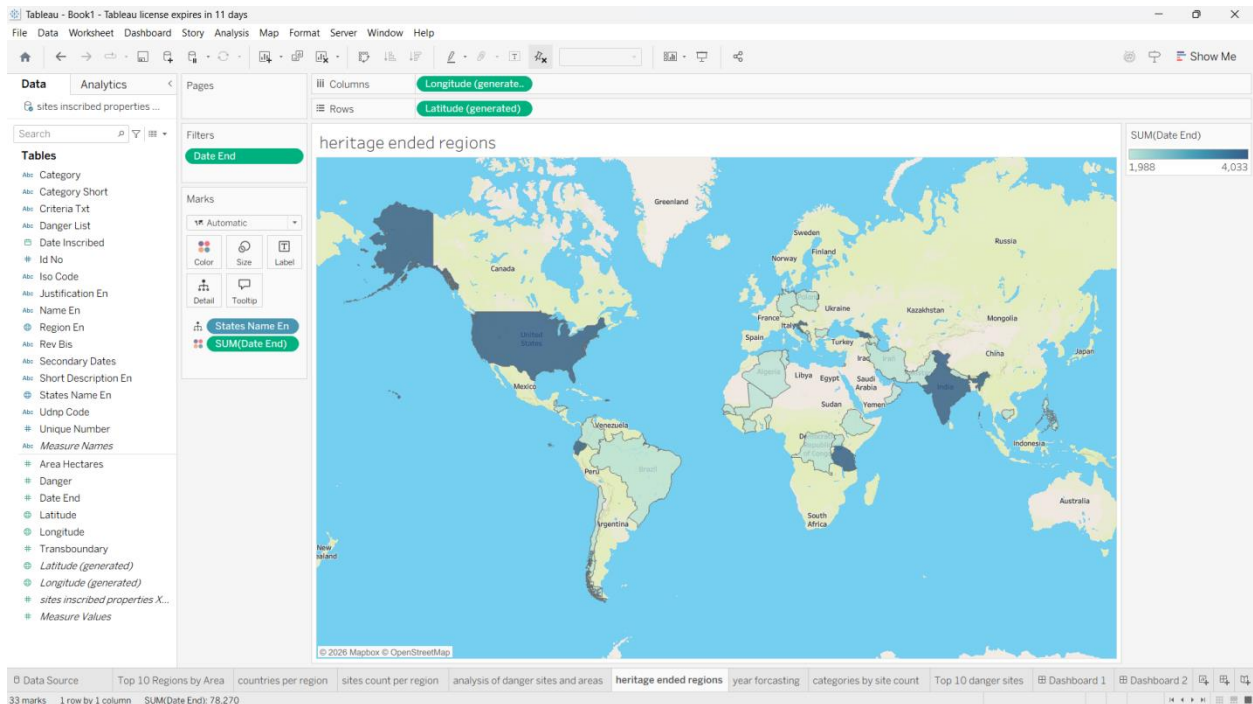
SHEET 5

HERITAGE ENDED REGIONS

- Create a worksheet titled Heritage Ended Regions, double click on States Name En and change the Mark Cards type to Map.
- Drag Date End to Filters, select All Values and then on Special tab, choose Non-null values.
- Drag Date End to Color and States Name En to Label.
- Go to Map Background Layers, change the style to Normal or Outdoors and deselect any unnecessary layers such as land borders.

OUTPUT

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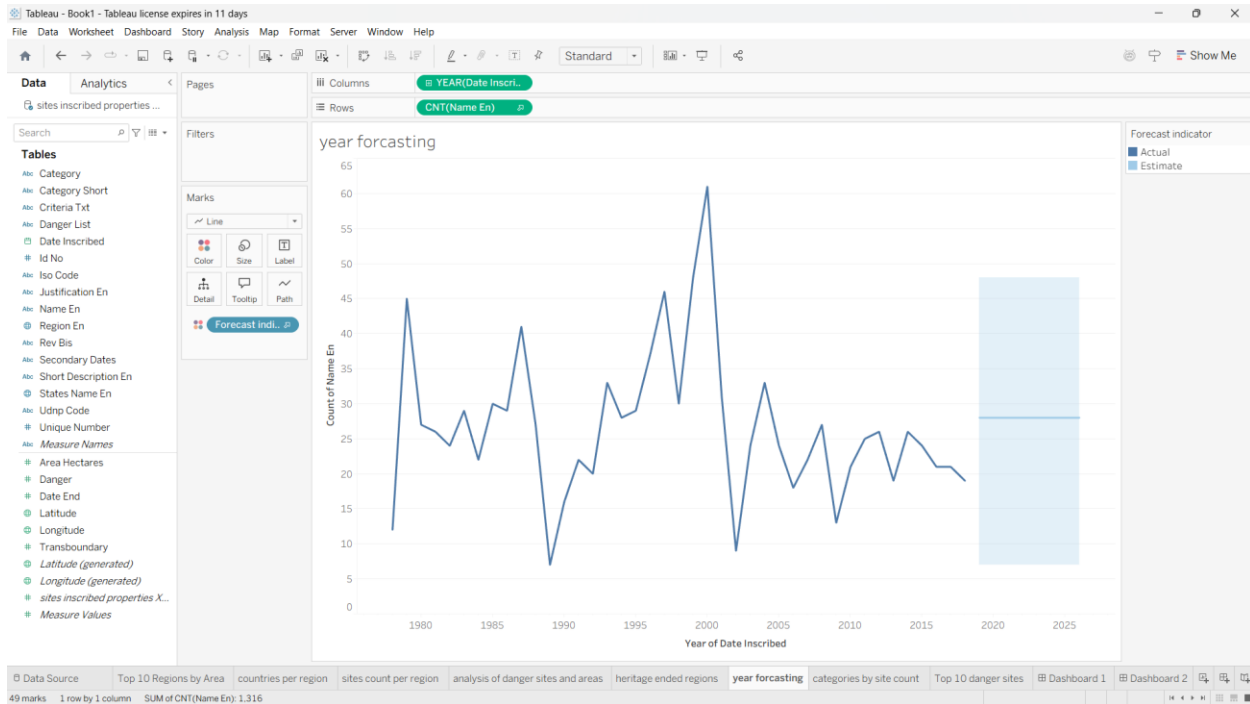
SHEET 6

YEAR FORECASTING

- Make the worksheet for year forecasting, move Date Inscribed to Columns, and then right-click it to set it as the continuous Year.
- Move Name En to Rows, right-click it, go to Measure, and then make it Count to form a line graph.
- Go to the Analytics tab beside Data pane, and then drag Forecast on the line graph.
- Right-click the year axis at the bottom of the graph, and then set the orientation as Up

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OUTPUT



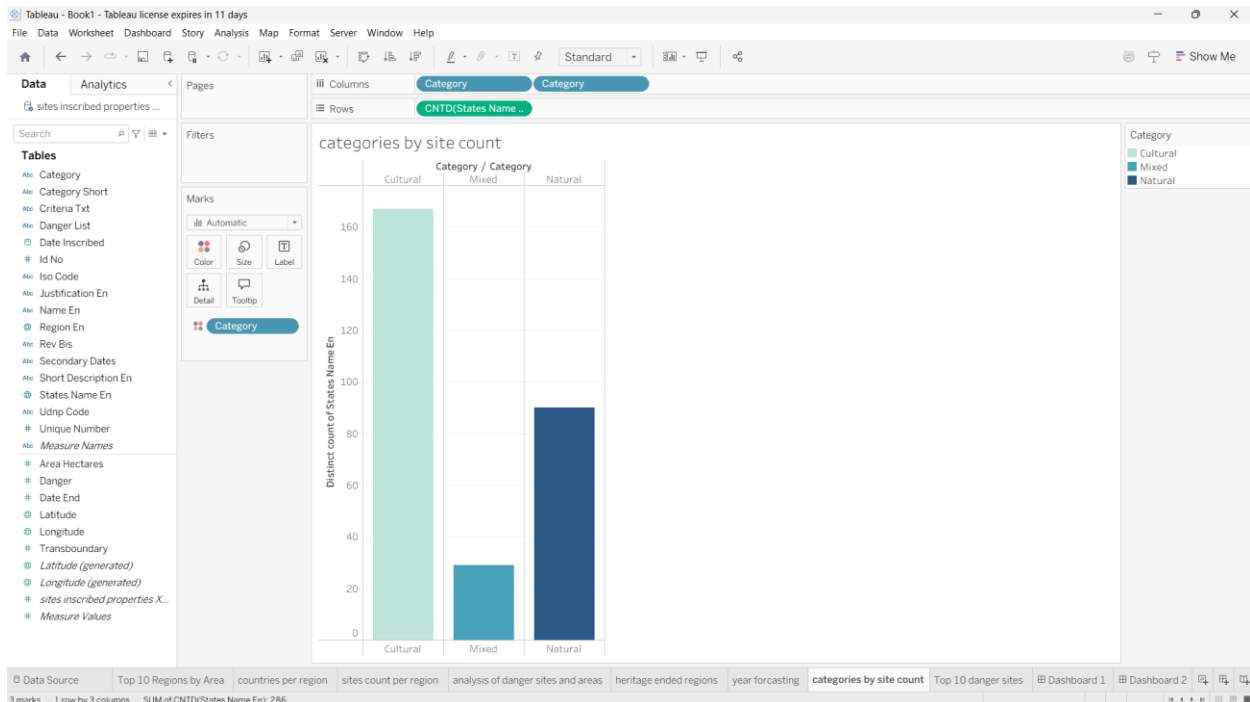
SHEET 7

CATEGORIES BY SITE COUNT

- Create new worksheet with name categories by site count and move the Category field into Columns.
- Drag States Name En field into Rows, right click on pill and change the measure to Count (Distinct).
- Move another copy of Category field into the Color tile on the Marks card.
- Move into Color > Edit colors and use the following color palette Light Sky Blue for Cultural, Medium Steel Blue for Mixed and Deep Cobalt Blue for Natural.
- Go to Size tile on the Marks card and use slider

OUTPUT

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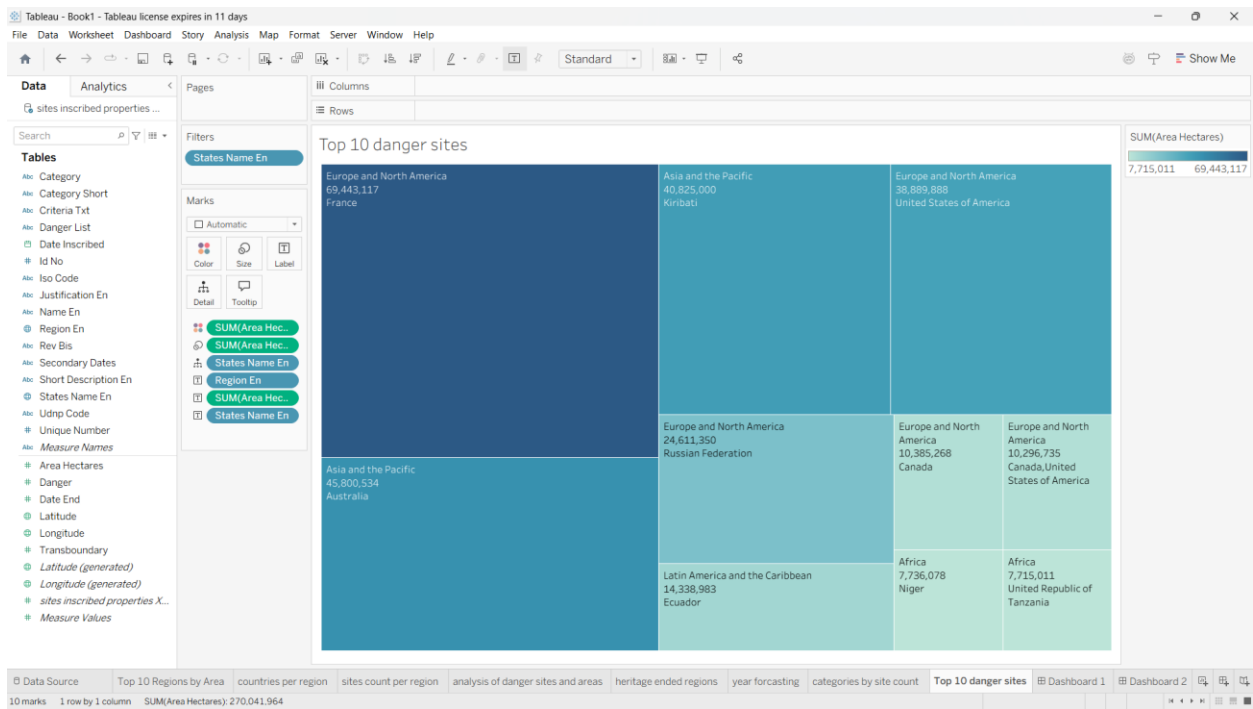
SHEET 8

TOP 10 DANGER SITES

- Create a new worksheet named Top 10 danger sites, pick Region, States Name and Area Hectares, and choose Treemap from Show Me.
- Place States Name on Label or Detail marks card in order to split your big regions into country blocks.
- Put States Name in the Filters shelf, select the Top tab, and configure it to Top 10 by SUM(Area Hectares).
- Drag Area Hectares and Region to the Label mark card in order to put all three variables in the text box.
- Press on Label, click on Text Editor, delete the default text, and rearrange the fields on three different rows starting with area, region, and states name.
- Click on Color, Edit Colors in order to change the gradient to blue or teal color.

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OUTPUT



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MAX MARKS	3 MARKS

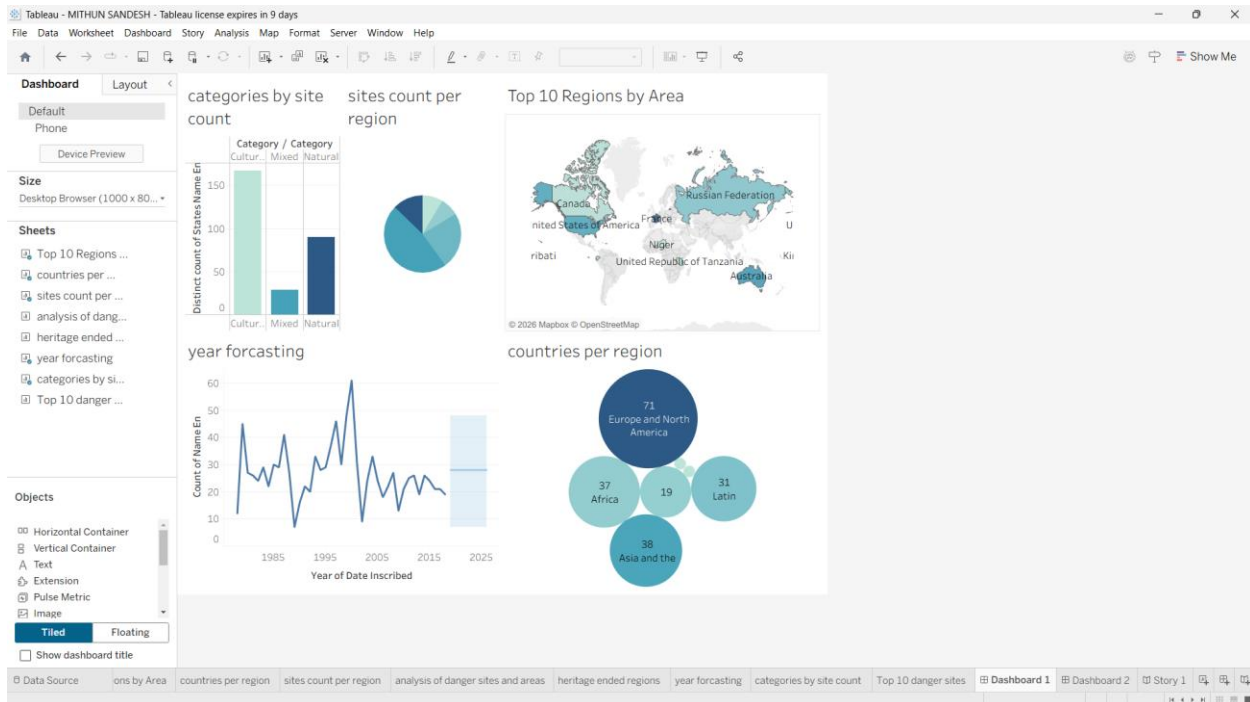
6.4 – Dashboard

DASHBOARD 1 :Heritage Overview Panel

- In New Dashboard, go to the Size tab and set it to Desktop Browser (1000 x 800). Also, lock the object type to Tiled.
- Arrange the objects based on category and site count and drag year forecast right below it to build a stack.
- Drag the Top 10 Regions by Area on to the right edge of the dashboard.
- Drag sites count per region to the left edge of the top map view and drag countries per region to the bottom edge of the top map view.
- Click the down arrow next to the auto-created color legends on the side of the dashboard and click Remove from Dashboard.

OUTPUT

Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

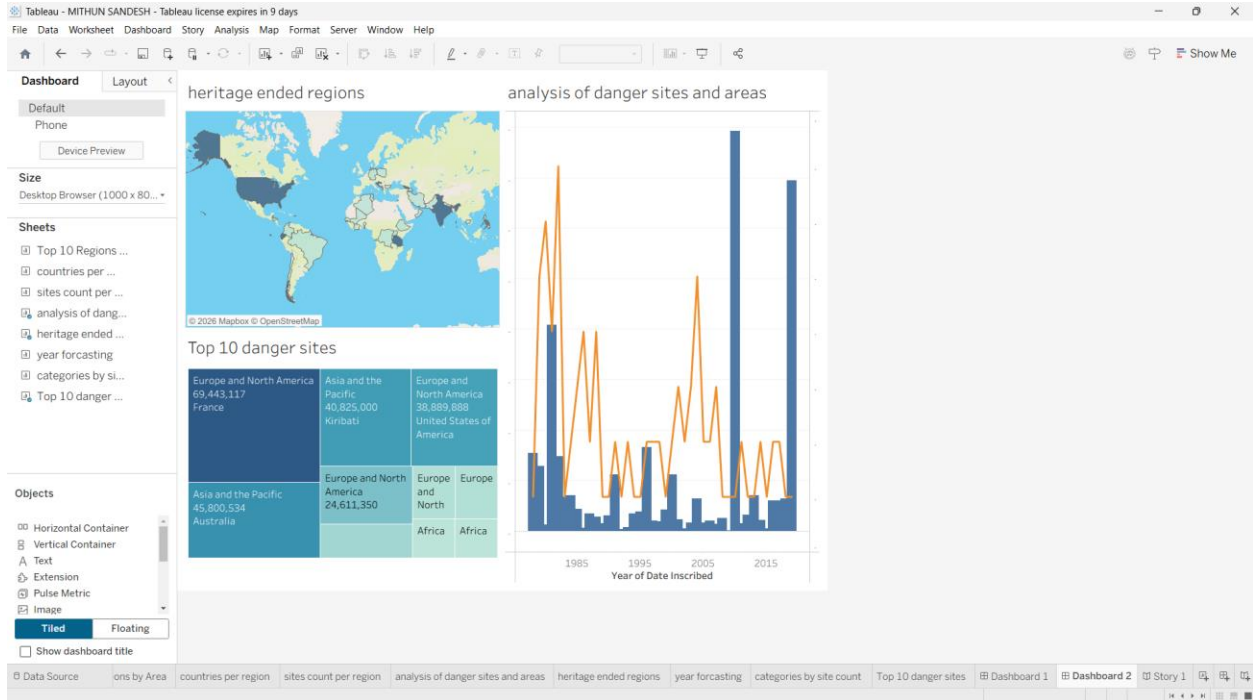


DASHBOARD 2: Danger Analysis Panel

- Click New Dashboard, and lock the object type as Tiled.
- Drag heritage ended regions on an empty Page , and drop analysis of danger sites and areas on its right edge to create two columns.
- Drag Top 10 danger sites and place it on the lower edge of heritage ended regions to form a grid in the bottom left corner.
- Place your cursor at the central vertical boundary of the chart till you see a resizing arrow, and then drag it to the left to expand the size of the line chart horizontally. ✨
- Click the drop-down arrow of any color legend which is not required on the side of the screen to delete it. Also double click the grey title bar to edit any text header.

OUTPUT

Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau



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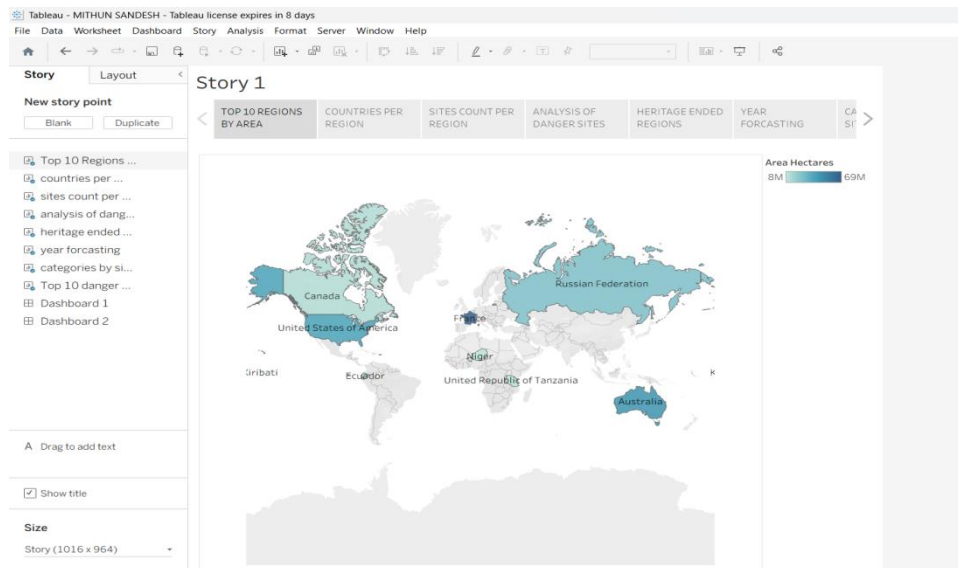
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TEAM ID	SWTID-2026-4638
PROJECT TITLE	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
MAX MARKS	3 MARKS

6.5 – Story

STORY 1

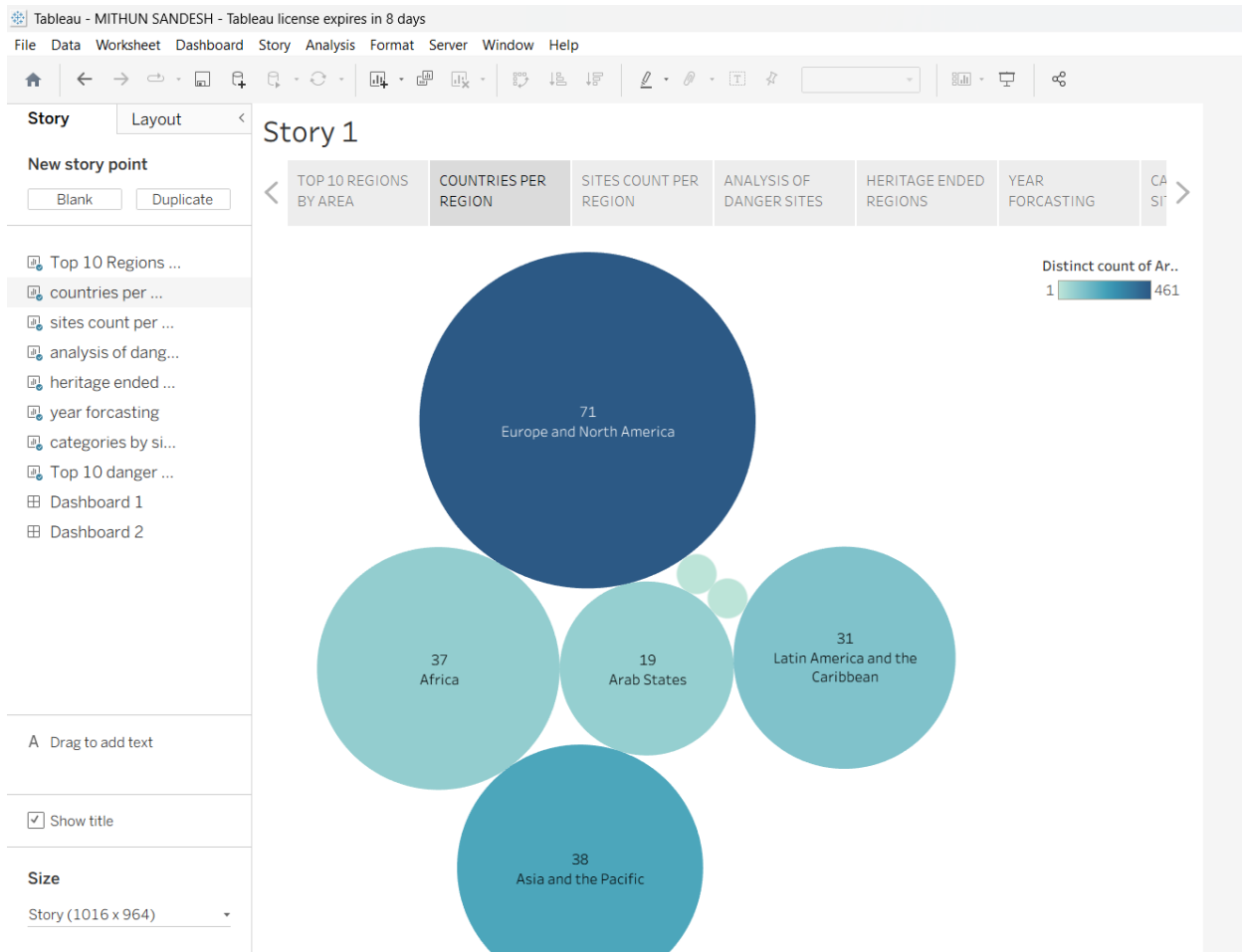
- Open new story in tableau
- Drag the charts and drop on sheets
- Add the captions to sheets
- On left pane, click blank for a new sheet



STORY 2

- On left pane,click blank for new sheet
- Drag charts and drop on sheet
- Add caption for sheets.

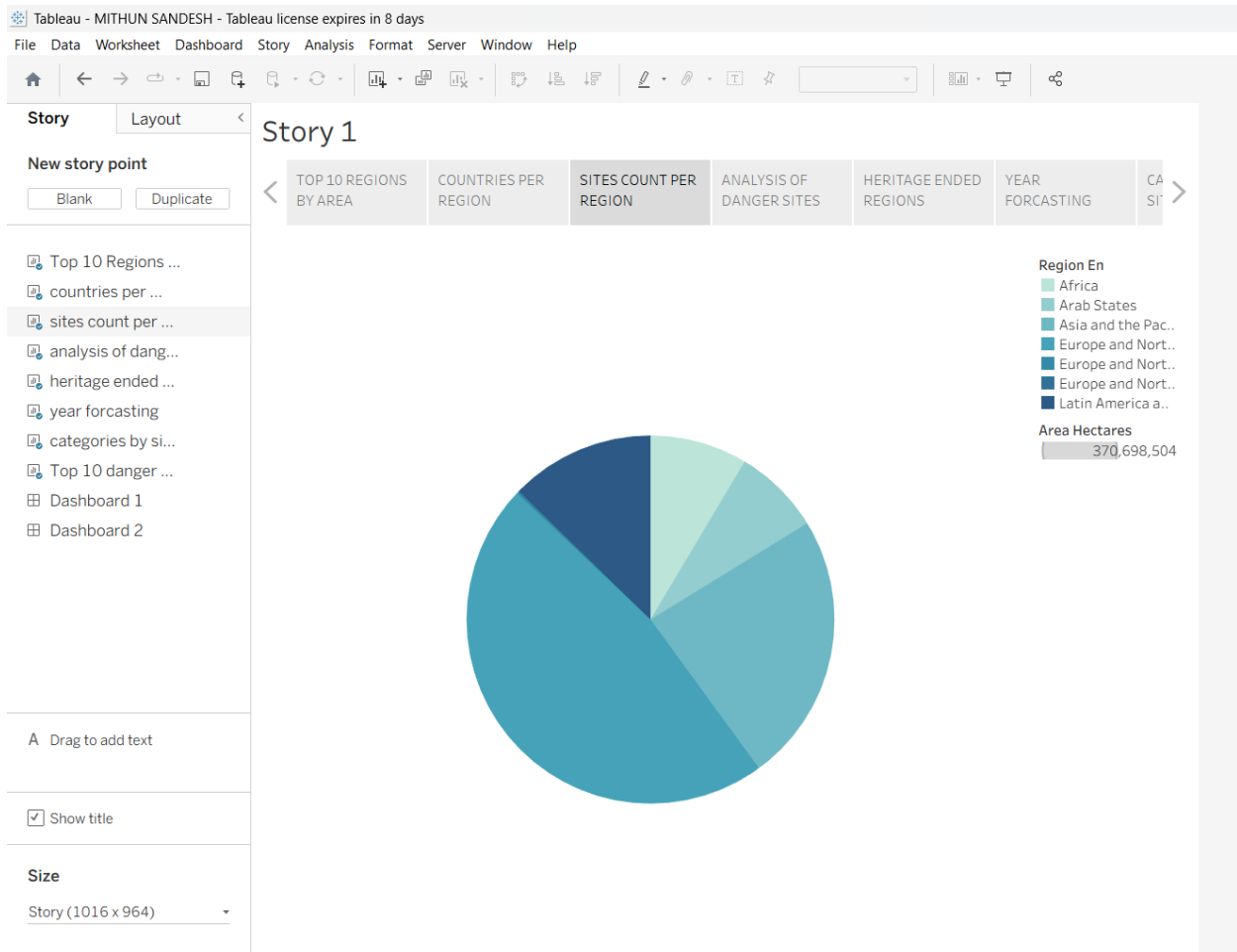
Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau



STORY 3

- On left pane,click blank for new sheet
- Drag charts and drop on sheet
- Add caption for sheets.

Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

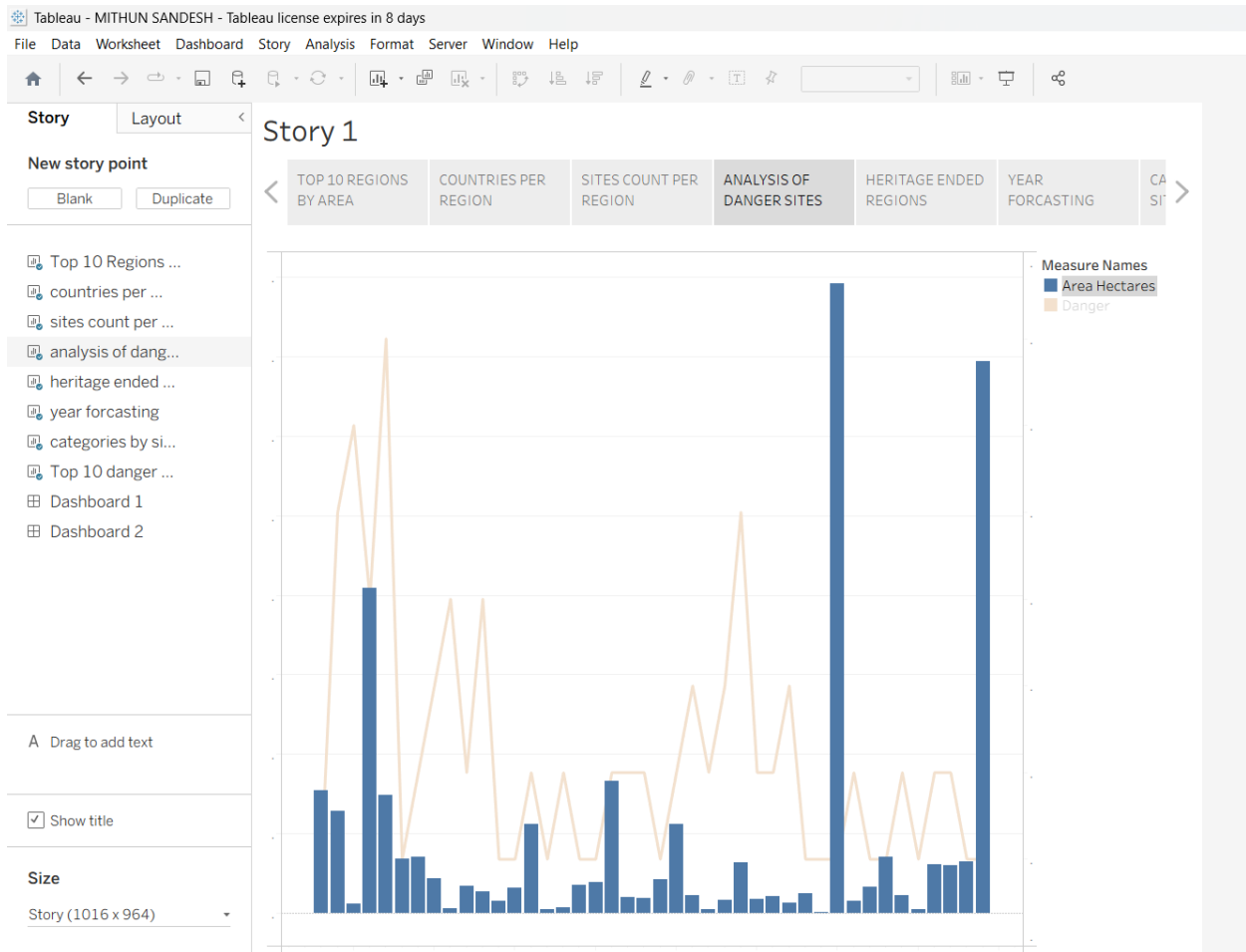


STORY 4

- On left pane,click blank for new sheet
- Drag charts and drop on sheet
- Add caption for sheets.

Heritage Treasures: An In-Depth Analysis of

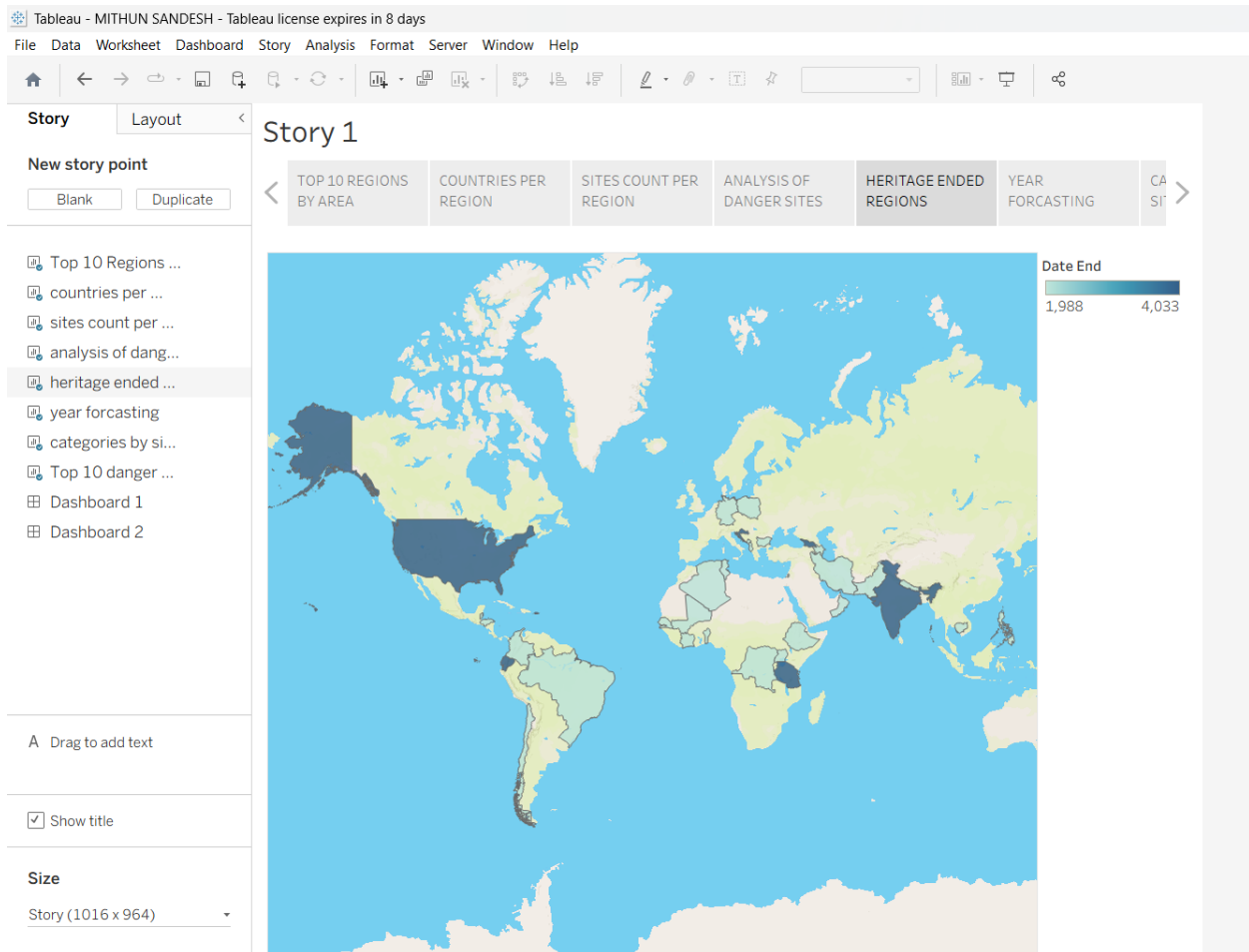
UNESCO World Heritage Sites in Tableau



STORY 5

- On left pane,click blank for new sheet
- Drag charts and drop on sheet
- Add caption for sheets.

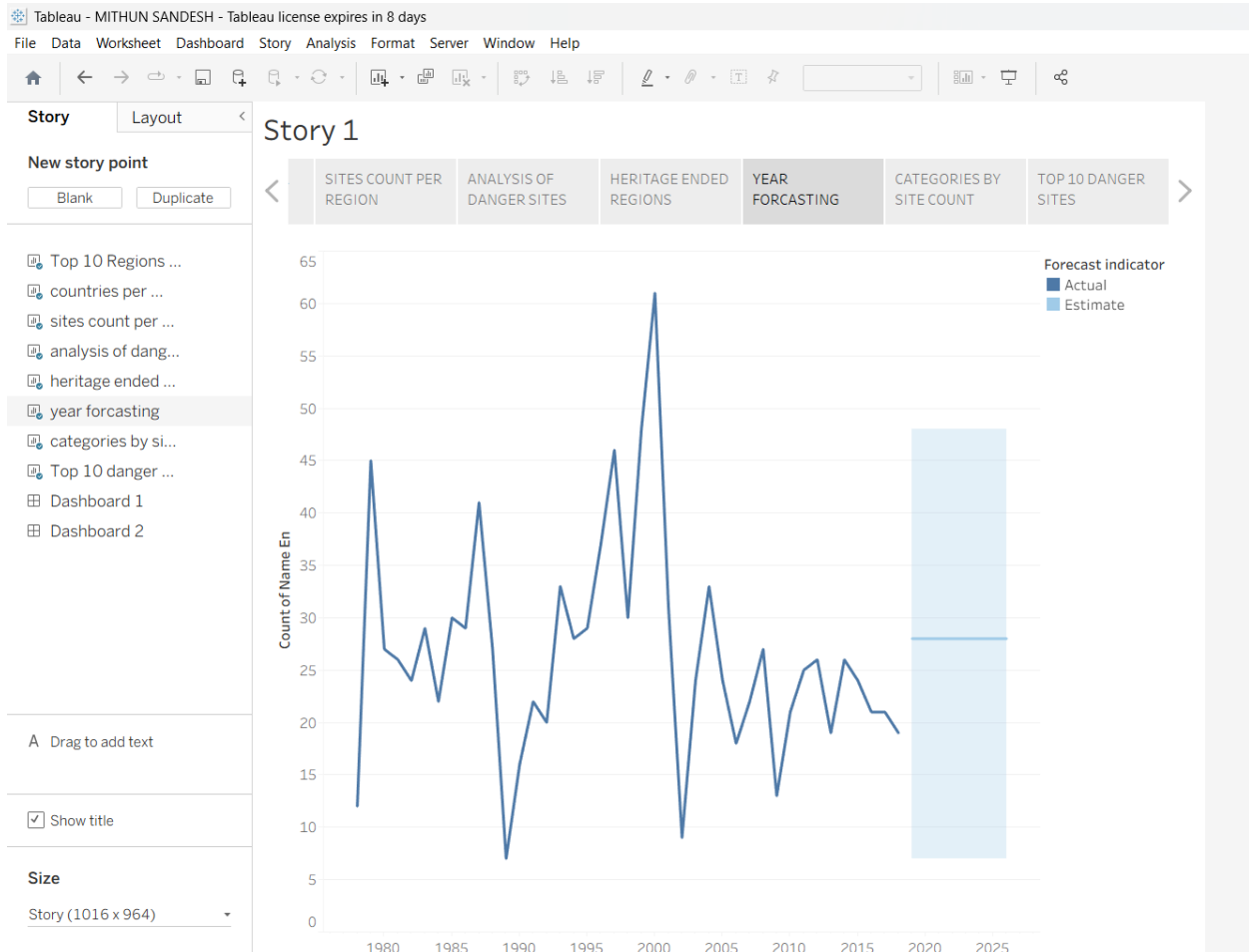
Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau



STORY 6

- On left pane,click blank for new sheet
- Drag charts and drop on sheet
- Add caption for sheets.

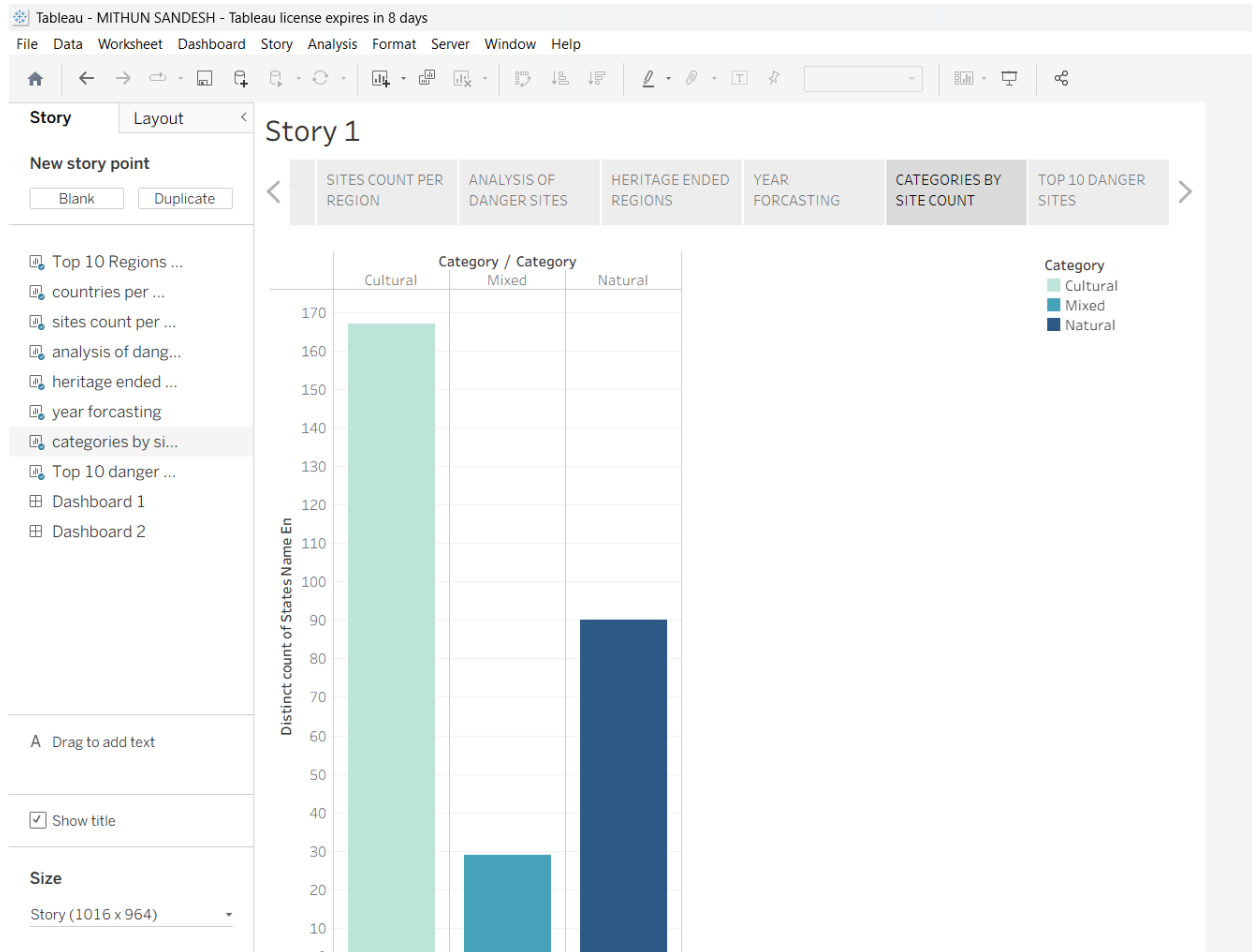
Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau



Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

STORY 7

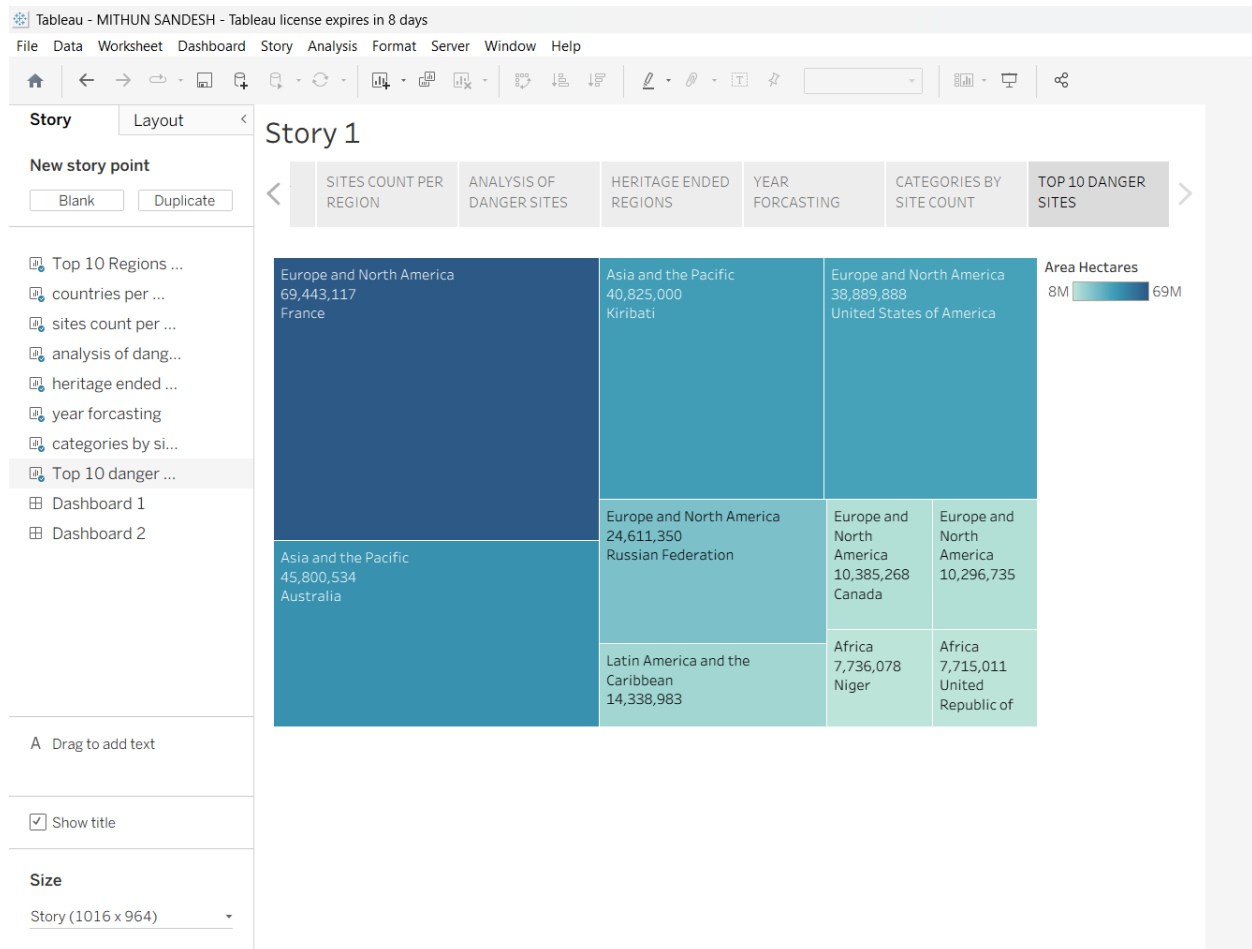
- On left pane,click blank for new sheet
- Drag charts and drop on sheet
- Add caption for sheets.



Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

STORY 8

- On left pane,click blank for new sheet
- Drag charts and drop on sheet
- Add caption for sheets.



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MAX MARKS	3 MARKS

6.6 – Publishing

DASHBOARD

Step 01: Connect to Tableau Public

- Navigate to the top menu in Tableau Desktop and select Server > Publish Workbook.
- If you are logged out, a prompt will automatically pop up asking for your details.
- Provide your Tableau Public login email and password, or create a new profile if needed.

Step 02: Name and Upload

- Once authentication is successful, a window will ask you to title your project.
- Select Save — this triggers Tableau to upload your workspace directly to the cloud.

Step 03: Check Your Online Visualization

- As soon as the transfer finishes, your default web browser will launch to display the dashboard live on your web profile.

STORY

Step 01: Connect to Your Tableau Public Profile

- Inside the Tableau Desktop application, navigate up to the top menu bar and select Server and Publish Workbook.
- If you aren't currently authenticated, a sign-in dialogue box will pop up on your screen.
- Key in your existing account details or register for a new profile if you don't have one.

Step 02: Title and Upload Your Project

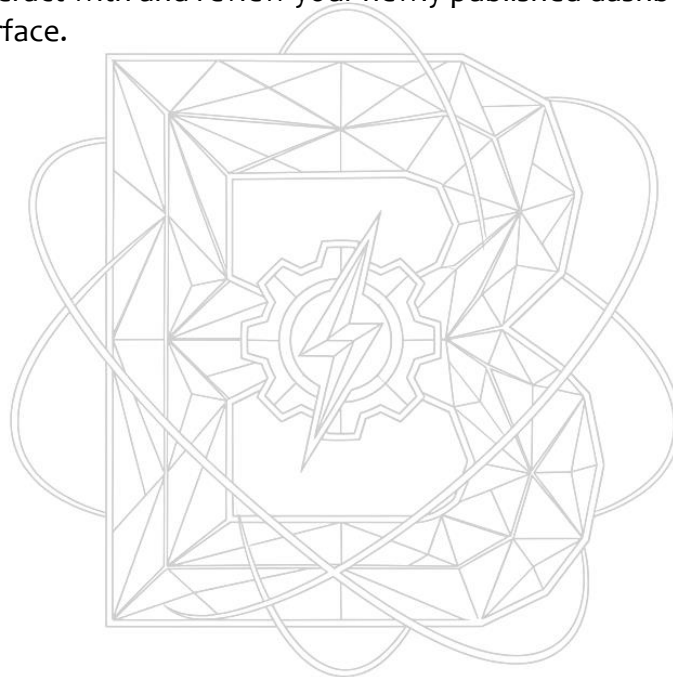
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- Once you are logged in, a window will appear prompting you to give your workbook a specific name.
- Press the Save button — Tableau will immediately begin pushing your active project files up to your cloud profile.

Step 03: Inspect Your Live story

- As soon as the upload completes, your web browser will automatically open up your personal profile page.
- You can now interact with and review your newly published dashboard or story directly on the web interface.



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MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

Amount of Data Loaded

The foundation of this Tableau analysis relies on a comprehensive dataset capturing hundreds of cultural, natural, and mixed UNESCO World Heritage sites spanning multiple decades across the globe. Loading this vast amount of data involves compiling extensive spatial coordinates (latitude and longitude)

Utilization of Data Filters

To transform this massive repository of global information into digestible, actionable insights, the strategic utilization of data filters is paramount. Tableau’s interactive filtering capabilities allow users to dynamically slice through the global dataset,

Number of Visualizations / Graphs

To tell a holistic story of the world's heritage treasures, the Tableau workbook deploys a carefully curated array of interconnected visualizations and graphs distributed across an interactive layout. Rather than relying on a single complex view, the analysis breaks down.

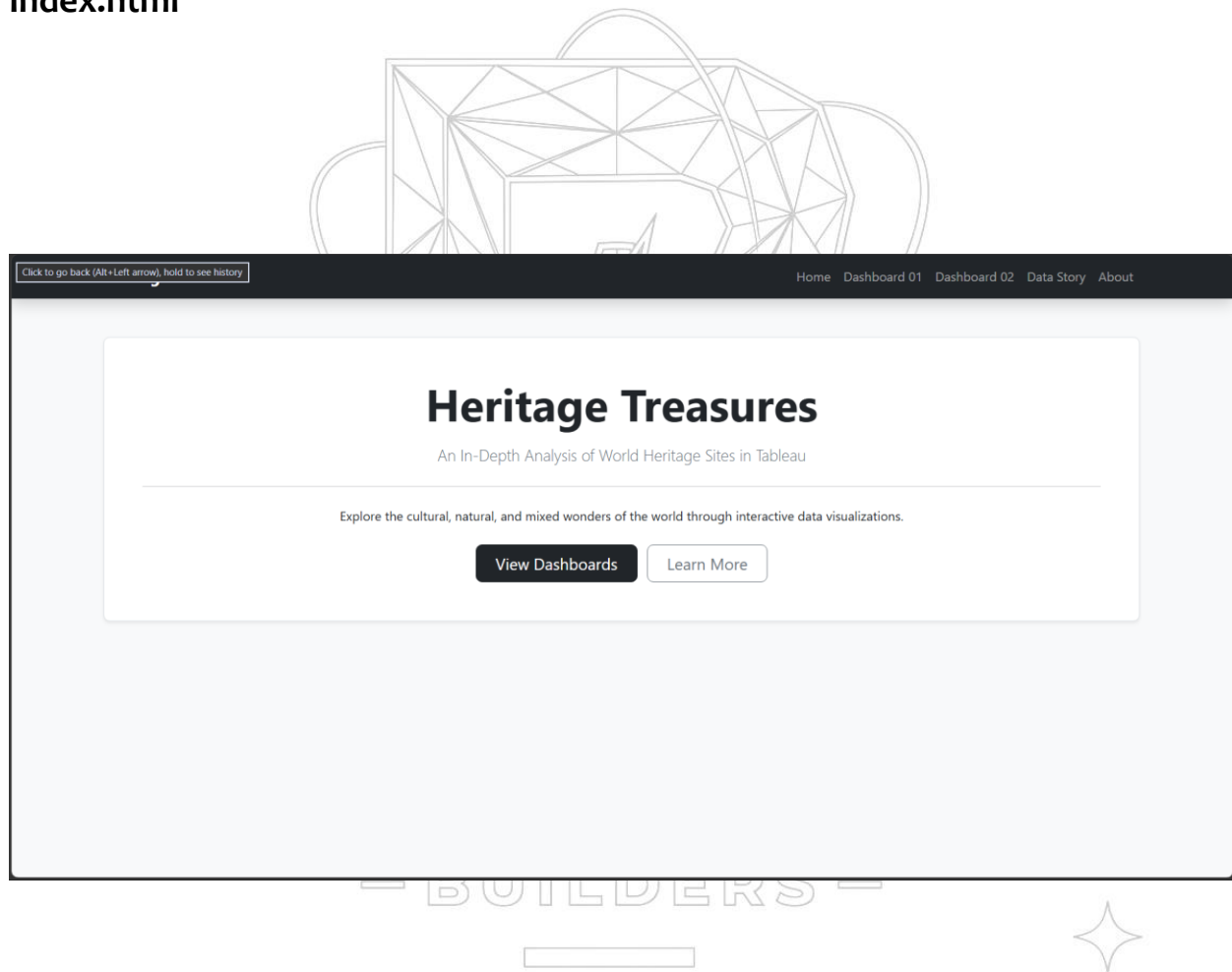
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CHAPTER – VIII

RESULTS

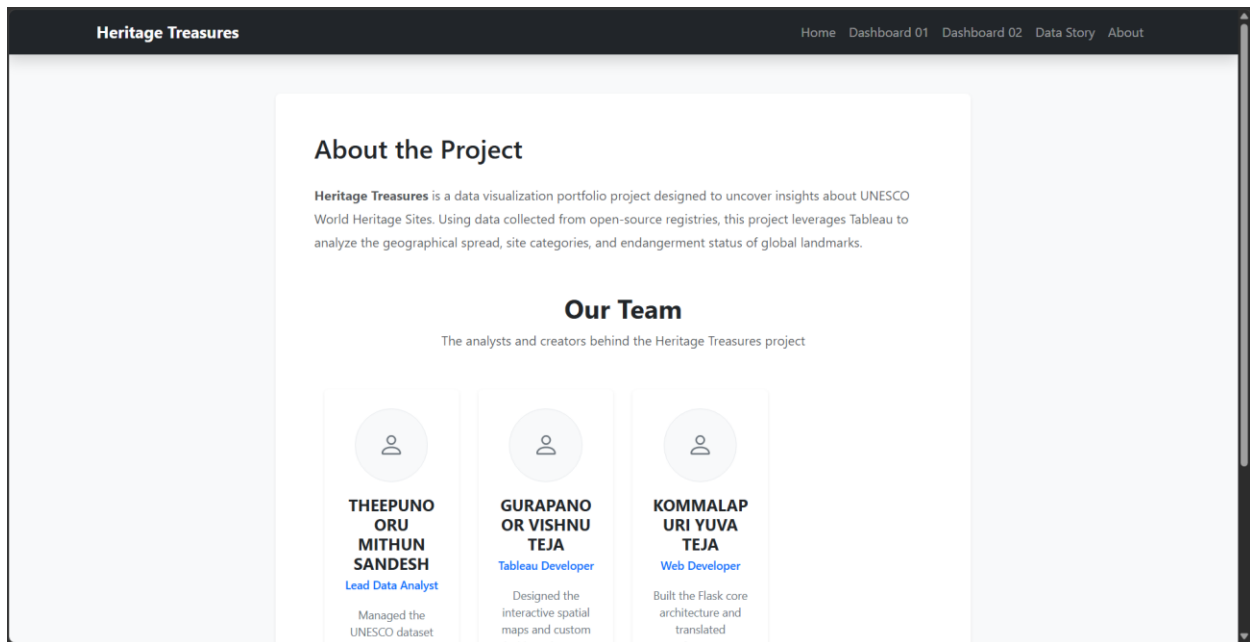
8.1 – Output Screenshots

Index.html

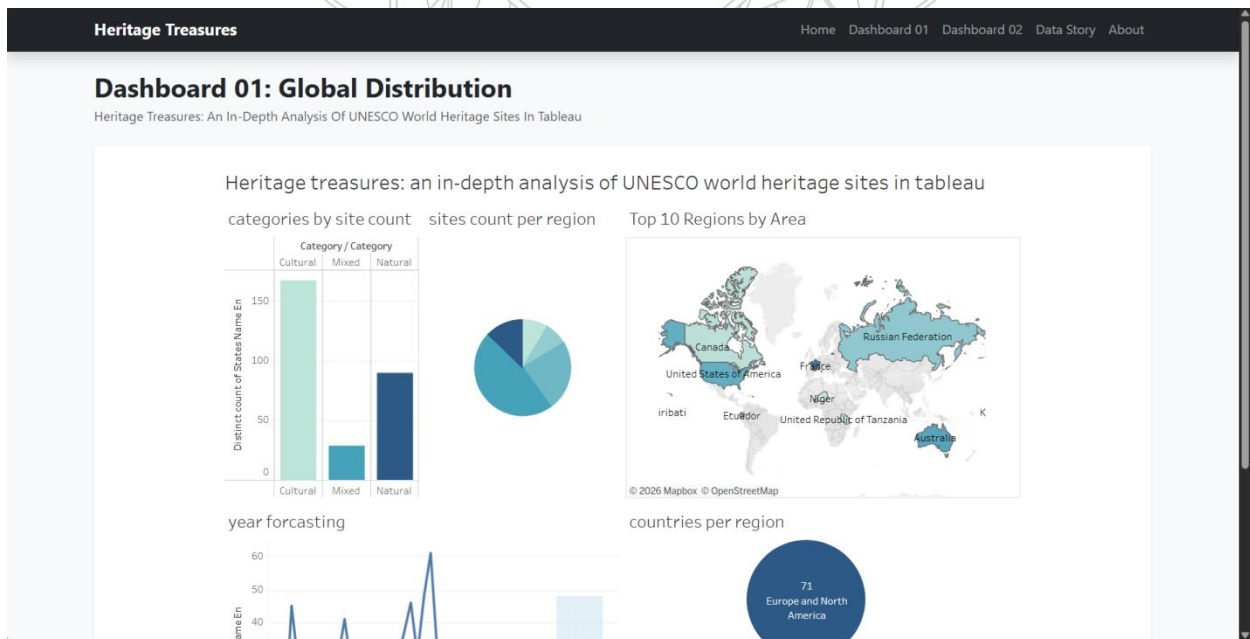


Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

About.html

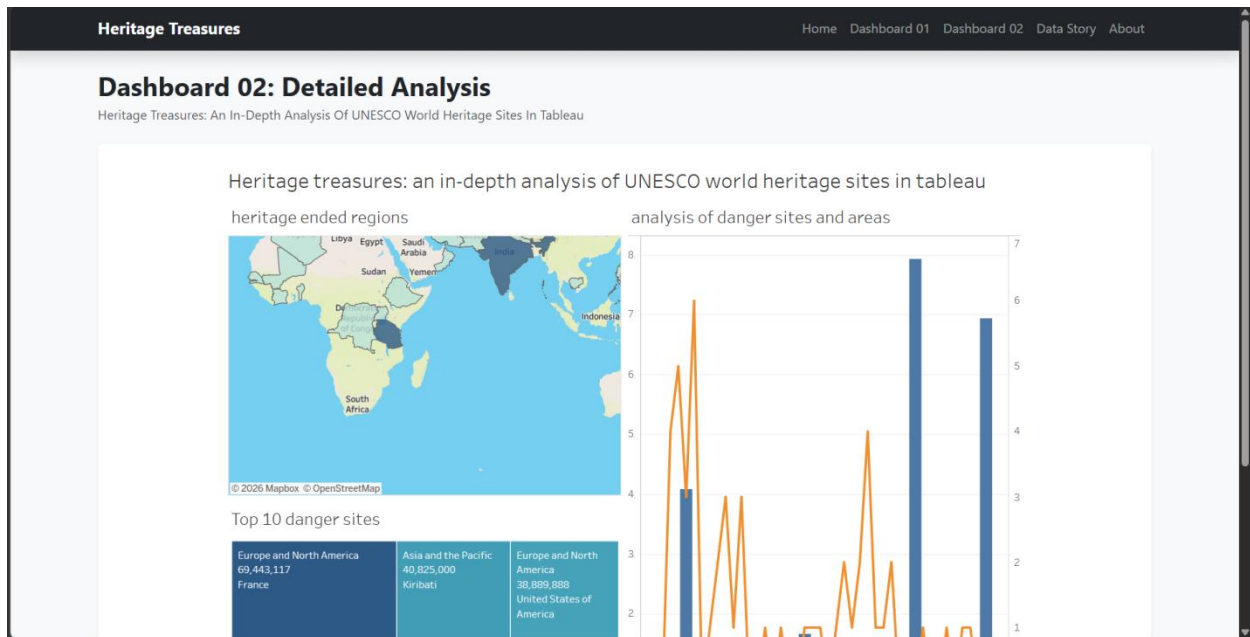


Dashboard 01

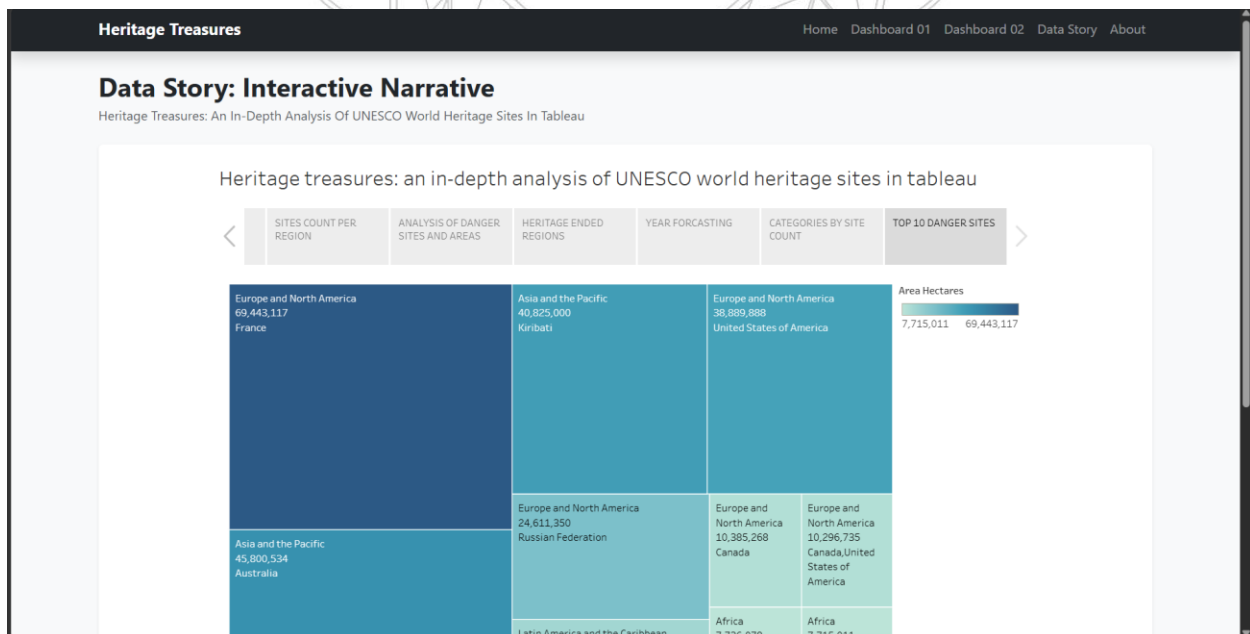


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Dashboard 02



Story



Heritage Treasures: An In-Depth Analysis of

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CHAPTER – IX

ADVANTAGES & DISADVANTAGES

9.1 - Advantages

The "Heritage Treasures" platform offers significant advantages over traditional data systems, providing an efficient, highly interactive workspace that transforms how researchers explore global heritage metadata. A primary benefit is the integration of advanced spatial analytics with traditional business intelligence tools. By replacing static text sheets with interactive maps, users can instantly discover spatial clusters and track regional trends that would remain hidden in standard spreadsheets, greatly accelerating the data discovery process. Another major advantage is the user-friendly, self-service design of the dashboards. By utilizing intuitive parameters, visual dropdowns, and responsive timeline sliders, the system allows non-technical users to run complex, multi-dimensional queries easily without needing to write SQL scripts or learn complicated GIS software. This accessibility bridges the gap between technical data layers and domain experts, enabling conservationists and policy advocates to extract clear, data-backed insights independently. Additionally, the platform's data model is optimized for high performance and easy maintenance. Built on the advanced Tableau Hyper engine, the workbook executes complex calculations and updates visual layouts in real time, ensuring a smooth, responsive user experience. The structured, logical relationships layer also allows the dataset to be updated quickly when UNESCO releases new records, creating a scalable, sustainable analytical tool that provides long-term value for academic research and global preservation advocacy.

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9.2 - Disadvantages

While the "Heritage Treasures" platform delivers powerful analytical tools, it has certain limitations shaped by its underlying data sources and software environment. A primary limitation stems from the project's dependency on the completeness and frequency of the official UNESCO source registries. If the source files contain delayed reporting from specific member nations or missing geographic coordinates for older, remote sites, the dashboards will reflect these gaps. The platform describes data trends but cannot automatically fix administrative data inconsistencies at the source level. Another limitation is that Tableau Desktop is primarily designed for structured data analysis rather than managing open-ended textual narratives. The platform excels at tracking quantitative metrics—such as site counts, regional density, and numeric risk indexes—but it cannot easily index or analyze long, unstructured text histories detailing a site's cultural background or legal battles. Users seeking deep, qualitative historical descriptions must consult external text records, as the dashboards are optimized for visual and statistical trends. Finally, the workbook relies on a desktop application environment, meaning that full interactive performance requires a computer that meets specific hardware and software prerequisites, including dedicated graphics processing and Tableau 2025.3.1 or later. If the platform is deployed on older, underpowered machines or exported to basic static formats like PDF or PowerPoint, it loses its core interactive capabilities—such as cross-filtering, dynamic parameters, and responsive tooltips—restricting users to a non-interactive view of the data.

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CHAPTER – X

CONCLUSION

10.1 - Conclusion

The "Heritage Treasures" data analytics project successfully demonstrates how modern visual intelligence can transform complex, fragmented global heritage data into clear, actionable insights. By using Tableau Desktop 2025.3.1, the project moves past the limitations of traditional, text-heavy databases to establish a high-performance workspace for spatial and historical analysis. The final dashboards give researchers a powerful tool to track the distribution of UNESCO sites, isolate endangered historical zones, and monitor preservation trends across different eras and regions. From a technical standpoint, the project validates the use of optimized data models and agile software practices in data engineering. Combining a high-speed Hyper extract with flexible logical relationships and level-of-detail (LOD) calculations allowed the system to maintain fast, sub-second response times while preserving the integrity of the underlying source data. The clean user interface, consistent typography, and intentional color design ensure the platform is highly functional, professional, and optimized for academic research and evaluation. In summary, this project highlights the essential role that data science and interactive visualization play in supporting global cultural preservation. Providing an intuitive, data-driven command center equips academic researchers, site managers, and policy advocates with the empirical evidence needed to champion conservation efforts. The completed platform stands as a robust, university-grade development success, effectively bridging the gap between raw archival data and impactful, visual storytelling for global heritage advocacy.

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CHAPTER – XI

FUTURE SCOPE

11.1 - Future Scope

The current implementation of the "Heritage Treasures" project provides a strong foundation for several exciting future developments and technical expansions. A primary goal for future iterations is to transition the standalone desktop workbook into a live, cloud-hosted dashboard via Tableau Cloud or Tableau Server. This upgrade will enable web-based, collaborative research workflows, allowing international conservation teams to access the platform simultaneously, share customized views, and receive automated data updates via live API connections to the UNESCO digital registries. Another key opportunity for growth is the integration of real-time environmental data streams into the spatial dashboards. By connecting the platform to external web services, future versions could overlay live climate tracking metrics—such as rising sea levels, regional deforestation rates, and active wildfire zones—directly onto the heritage maps. This expansion would transform the system from a descriptive reporting tool into a proactive, predictive risk model, helping organizations identify and protect threatened historic assets before irreversible damage occurs.

Finally, the platform's analytical capabilities can be enhanced by incorporating advanced predictive models, such as machine learning algorithms written in Python or R and connected through Tableau's external services integration (TabPy). These models could analyze historical funding distributions, localized population growth, and regional stability indicators to project future risk scores for specific site categories. This data-driven forecasting would provide global policy advocates with invaluable insights, helping them optimize funding allocations and shape long-term heritage preservation strategies.

Heritage Treasures: An In-Depth Analysis of

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CHAPTER – XII

APPENDIX

12.1 – Source Code

App.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

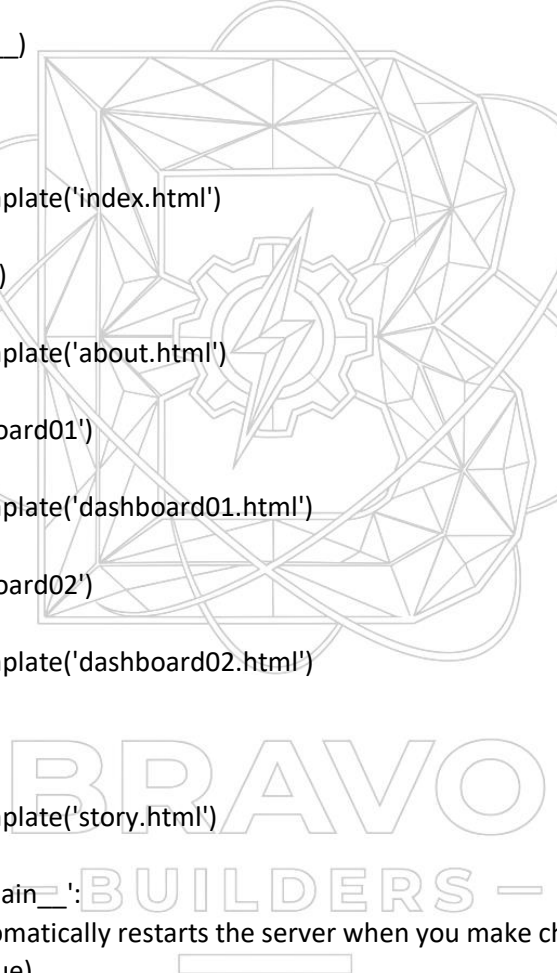
@app.route('/about')
def about():
    return render_template('about.html')

@app.route('/dashboard01')
def dashboard01():
    return render_template('dashboard01.html')

@app.route('/dashboard02')
def dashboard02():
    return render_template('dashboard02.html')

@app.route('/story')
def story():
    return render_template('story.html')

if __name__ == '__main__':
    # debug=True automatically restarts the server when you make changes
    app.run(debug=True)
```



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Navbar.html

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark mb-4 shadow">
  <div class="container">
    <a class="navbar-brand fw-bold" href="{{ url_for('index') }}">Heritage Treasures</a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navbarNav">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarNav">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item"><a class="nav-link" href="{{ url_for('index') }}">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="{{ url_for('dashboard01') }}">Dashboard
01</a></li>
        <li class="nav-item"><a class="nav-link" href="{{ url_for('dashboard02') }}">Dashboard
02</a></li>
        <li class="nav-item"><a class="nav-link" href="{{ url_for('story') }}">Data Story</a></li>
        <li class="nav-item"><a class="nav-link" href="{{ url_for('about') }}">About</a></li>
      </ul>
    </div>
  </div>
</nav>
```

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Heritage Treasures | Home</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>body { background-color: #f8f9fa; }</style>
</head>
<body>
  {% include 'navbar.html' %}
```

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```
<div class="container mt-5">
  <div class="p-5 text-center bg-white rounded-3 shadow-sm border">
    <h1 class="display-4 fw-bold text-dark">Heritage Treasures</h1>
    <p class="lead text-secondary mt-3">An In-Depth Analysis of World Heritage Sites in
Tableau</p>
    <hr class="my-4">
    <p>Explore the cultural, natural, and mixed wonders of the world through interactive
data visualizations.</p>
    <div class="mt-4">
      <a class="btn btn-dark btn-lg px-4 me-2" href="{{ url_for('dashboard01') }}">View
Dashboards</a>
      <a class="btn btn-outline-secondary btn-lg px-4" href="{{ url_for('about') }}">Learn
More</a>
    </div>
  </div>
</div>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Dashboard 01

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Heritage Treasures | Dashboard 01</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    body {
      background-color: #f8f9fa;
    }
    /* Forces a stable container block so the Tableau JS engine doesn't read 0px width */
    .tableau-responsive-container {
```

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```

width: 100%;
max-width: 1000px;
margin: 0 auto;
overflow-x: auto;
-webkit-overflow-scrolling: touch;
}
</style>
</head>
<body>
{% include 'navbar.html' %}

<div class="container my-4">
  <div class="mb-4">
    <h2 class="fw-bold text-dark mb-1">Dashboard 01: Global Distribution</h2>
    <p class="text-muted text-capitalize">Heritage treasures: an in-depth analysis of UNESCO
world heritage sites in tableau</p>
  </div>

  <div class="card shadow-sm border-0">
    <div class="card-body p-3 bg-white rounded">
      <div class="tableau-responsive-container">

        <div class='tableauPlaceholder' id='viz1782282524358' style='position: relative;
width: 100%;'>
          <noscript>
            <a href='#'>
              <img alt='Heritage treasures: an in-depth analysis of UNESCO world heritage
sites in tableau '
src='https://public.tableau.com/static/images/DA/DASHBOARD1-
Heritagetreasuresanin-
depthanalysisofUNESCOworldheritagesitesintableau/Dashboard1/1_rss.png'
style='border: none' />
            </a>
          </noscript>
          <object class='tableauViz' style='display:none;'>
            <param name='host_url' value='https%3A%2F%2Fpublic.tableau.com%2F' />
            <param name='embed_code_version' value='3' />
            <param name='site_root' value='' />
            <param name='name' value='DASHBOARD1-Heritagetreasuresanin-
depthanalysisofUNESCOworldheritagesitesintableau/Dashboard1' />
            <param name='tabs' value='no' />
            <param name='toolbar' value='yes' />
            <param name='static_image'

```

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value='https://public.tableau.com/static/images/DA/DASHBOARD1-HeritageTreasuresAnIn-

depthanalysisofUNESCOWorldHeritageSitesInTableau/Dashboard1/1.png' />

<param name='animate_transition' value='yes' />

<param name='display_static_image' value='yes' />

<param name='display_spinner' value='yes' />

<param name='display_overlay' value='yes' />

<param name='display_count' value='yes' />

<param name='language' value='en-US' />

</object>

</div>

<script type='text/javascript'>

// Wrapping inside window.onload prevents layout calculation errors

window.onload = function() {

var divElement = document.getElementById('viz1782282524358');

var vizElement = divElement.getElementsByTagName('object')[0];

// Safety check: if offsetWidth is 0, fall back to window width or a default 1000px

var calculatedWidth = divElement.offsetWidth || window.innerWidth || 1000;

if (calculatedWidth > 800) {

vizElement.style.width='1000px';

vizElement.style.height='827px';

} else if (calculatedWidth > 500) {

vizElement.style.width='1000px';

vizElement.style.height='827px';

} else {

vizElement.style.width='100%';

vizElement.style.height='1577px';

}

var scriptElement = document.createElement('script');

scriptElement.src = 'https://public.tableau.com/javascripts/api/viz_v1.js';

vizElement.parentNode.insertBefore(scriptElement, vizElement);

};

</script>

</div>

</div>

</div>

</div>

<script

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```
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Dashboard 02

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Heritage Treasures | Dashboard 02</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    body {
      background-color: #f8f9fa;
    }
    /* Forces a stable container block so the Tableau JS engine doesn't read 0px width */
    .tableau-responsive-container {
      width: 100%;
      max-width: 1000px;
      margin: 0 auto;
      overflow-x: auto;
      -webkit-overflow-scrolling: touch;
    }
  </style>
</head>
<body>
  {% include 'navbar.html' %}

  <div class="container my-4">
    <div class="mb-4">
      <h2 class="fw-bold text-dark mb-1">Dashboard 02: Detailed Analysis</h2>
      <p class="text-muted text-capitalize">Heritage treasures: an in-depth analysis of UNESCO
world heritage sites in tableau</p>
    </div>

    <div class="card shadow-sm border-0">
```

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```

<div class="card-body p-3 bg-white rounded">
  <div class="tableau-responsive-container">

    <div class='tableauPlaceholder' id='viz1782282637003' style='position: relative;
width: 100%;'>
      <noscript>
        <a href='#'>
          <img alt='Heritage treasures: an in-depth analysis of UNESCO world heritage
sites in tableau '
src='https://public.tableau.com/static/images/DA/DASHBOARD2-
Heritagetreasuresanin-
depthanalysisofUNESCOworldheritagesitesintableau/Dashboard2/1_rss.png'
style='border: none' />
        </a>
      </noscript>
      <object class='tableauViz' style='display:none;'>
        <param name='host_url' value='https%3A%2F%2Fpublic.tableau.com%2F' />
        <param name='embed_code_version' value='3' />
        <param name='site_root' value='' />
        <param name='name' value='DASHBOARD2-Heritagetreasuresanin-
depthanalysisofUNESCOworldheritagesitesintableau/Dashboard2' />
        <param name='tabs' value='no' />
        <param name='toolbar' value='yes' />
        <param name='static_image'
value='https://public.tableau.com/static/images/DA/DASHBOAR
D2-Heritagetreasuresanin-
depthanalysisofUNESCOworldheritagesitesintableau/Dashboard2/1.png' />
        <param name='animate_transition' value='yes' />
        <param name='display_static_image' value='yes' />
        <param name='display_spinner' value='yes' />
        <param name='display_overlay' value='yes' />
        <param name='display_count' value='yes' />
        <param name='language' value='en-US' />
      </object>
    </div>

    <script type='text/javascript'>
      // Wrapping inside window.onload prevents layout calculation errors
      window.onload = function() {
        var divElement = document.getElementById('viz1782282637003');
        var vizElement = divElement.getElementsByTagName('object')[0];

        // Safety check: if offsetWidth is 0, fall back to window width or a default 1000px

```

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```
var calculatedWidth = divElement.offsetWidth || window.innerWidth || 1000;

if ( calculatedWidth > 800 ) {
  vizElement.style.width='1000px';
  vizElement.style.height='827px';
} else if ( calculatedWidth > 500 ) {
  vizElement.style.width='1000px';
  vizElement.style.height='827px';
} else {
  vizElement.style.width='100%';
  vizElement.style.height='977px';
}

var scriptElement = document.createElement('script');
scriptElement.src = 'https://public.tableau.com/javascripts/api/viz_v1.js';
vizElement.parentNode.insertBefore(scriptElement, vizElement);
};
</script>
</div>
</div>
</div>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Story.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Heritage Treasures | Data Story</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    body {
```

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```

background-color: #f8f9fa;
}
/* Forces a stable container block so the Tableau JS engine doesn't read 0px width */
.tableau-responsive-container {
width: 100%;
max-width: 1016px; /* Configured to match the story layout width */
margin: 0 auto;
overflow-x: auto;
-webkit-overflow-scrolling: touch;
}
</style>
</head>
<body>
{% include 'navbar.html' %}

<div class="container my-4">
  <!-- Page Header -->
  <div class="mb-4">
    <h2 class="fw-bold text-dark mb-1">Data Story: Interactive Narrative</h2>
    <p class="text-muted text-capitalize">Heritage treasures: an in-depth analysis of UNESCO
world heritage sites in tableau</p>
  </div>

  <!-- Interactive Visualization Card Container -->
  <div class="card shadow-sm border-0">
    <div class="card-body p-3 bg-white rounded">
      <div class="tableau-responsive-container">

        <!-- Tableau Embed Section Start -->
        <div class='tableauPlaceholder' id='viz1782282706944' style='position: relative;
width: 100%;'>
          <noscript>
            <a href='#'>
              <img alt='Heritage treasures: an in-depth analysis of UNESCO world heritage
sites in tableau '
src='https://public.tableau.com/static/images/ST/STORY-
HeritageTreasuresAnIn-
depthAnalysisOfUNESCOWorldHeritageSitesInTableau/Story1/1_rss.png' style='border:
none' />
            </a>
          </noscript>
          <object class='tableauViz' style='display:none;'>
            <param name='host_url' value='https://public.tableau.com' />

```

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```
<param name='embed_code_version' value='3' />
<param name='site_root' value='' />
<param name='name' value='STORY-Heritagetreasuresanin-
depthanalysisofUNESCOworldheritagesitesintableau&#47;Story1' />
<param name='tabs' value='no' />
<param name='toolbar' value='yes' />
<param name='static_image'
value='https:&#47;&#47;public.tableau.com&#47;static&#47;images&#47;ST&#47;STORY-
Heritagetreasuresanin-
depthanalysisofUNESCOworldheritagesitesintableau&#47;Story1&#47;1.png' />
<param name='animate_transition' value='yes' />
<param name='display_static_image' value='yes' />
<param name='display_spinner' value='yes' />
<param name='display_overlay' value='yes' />
<param name='display_count' value='yes' />
<param name='language' value='en-US' />
</object>
</div>

<script type='text/javascript'>
// Wrapping inside window.onload prevents layout calculation errors
window.onload = function() {
    var divElement = document.getElementById('viz1782282706944');
    var vizElement = divElement.getElementsByTagName('object')[0];

    // Sets dimensions safely after the browser finishes painting the card layout
    vizElement.style.width='1016px';
    vizElement.style.height='991px';

    var scriptElement = document.createElement('script');
    scriptElement.src = 'https://public.tableau.com/javascripts/api/viz_v1.js';
    vizElement.parentNode.insertBefore(scriptElement, vizElement);
};
</script>
<!-- Tableau Embed Section End -->

</div>
</div>
</div>
</div>

<!-- Bootstrap JS Bundle -->
<script
```

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```
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Style.css

```
body{
  background:#f5f5f5;
  font-family:Arial, Helvetica, sans-serif;
}
```

```
.hero{
  padding:100px 20px;
  background:white;
  text-align:center;
}
```

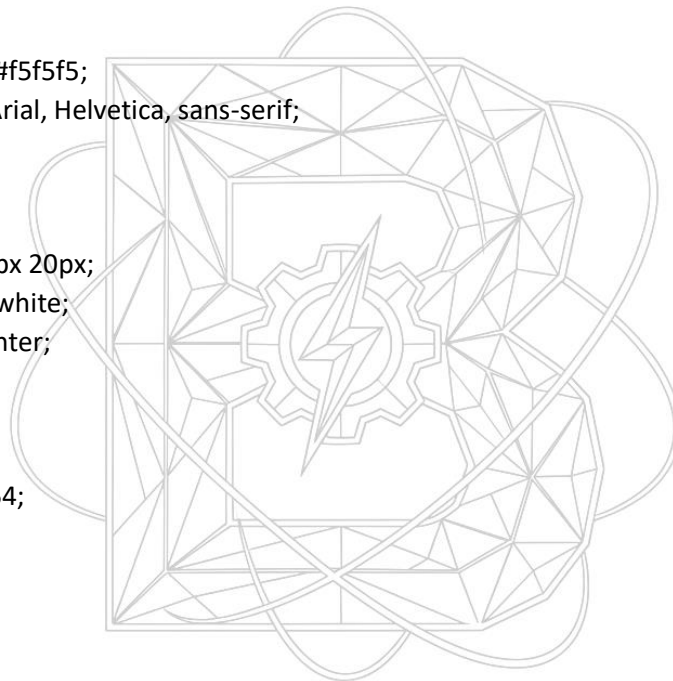
```
.hero h1{
  color:#198754;
}
```

```
.hero h4{
  color:#555;
}
```

```
footer{
  margin-top:60px;
}
```

```
iframe{
  border:none;
  width:100%;
  height:100%;
  border-radius:10px;
  box-shadow:0 0 10px rgba(0,0,0,.15);
}
```

```
.container{
```

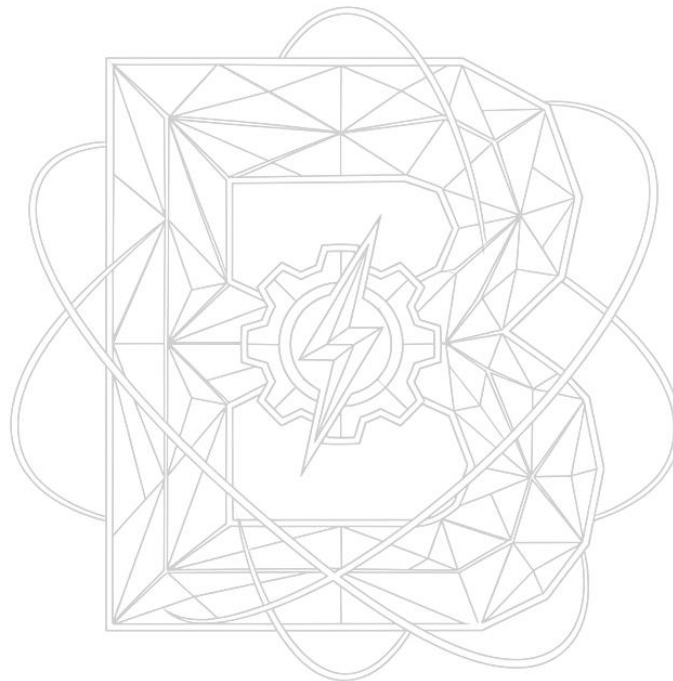


BRAVO
— BUILDERS —
—



Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

```
max-width:1200px;  
}
```



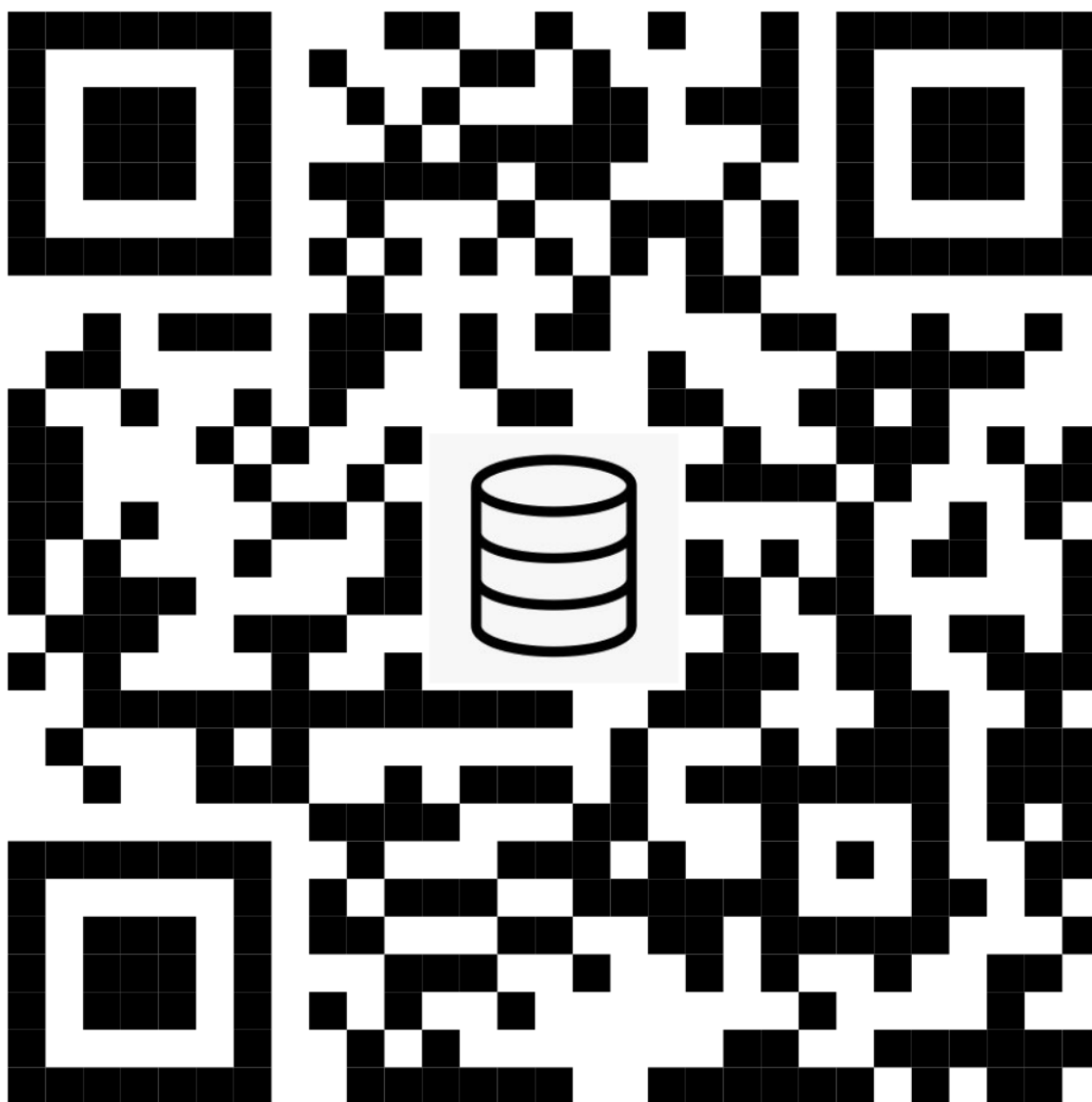
BRAVO
— BUILDERS —
—



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12.2 – Dataset Link

<https://www.kaggle.com/datasets/ujwalkandi/unesco-world-heritage-sites/data?select=whc-sites-2019.csv>



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12.3 – GitHub & Project Demo Link

<https://github.com/infinitycoders-046/SWTID-2026-4638-THEEPUNOORU-MITHUN-SANDESH>



Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

<https://drive.google.com/file/d/1LjNoKRbOI05rK5GExNuhvq5NhCXLv2Rv/view>

